Is this model mine? On stealing and defending machine learning models

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Baltic Energy and Cyber Security Forum 2025

CISPA
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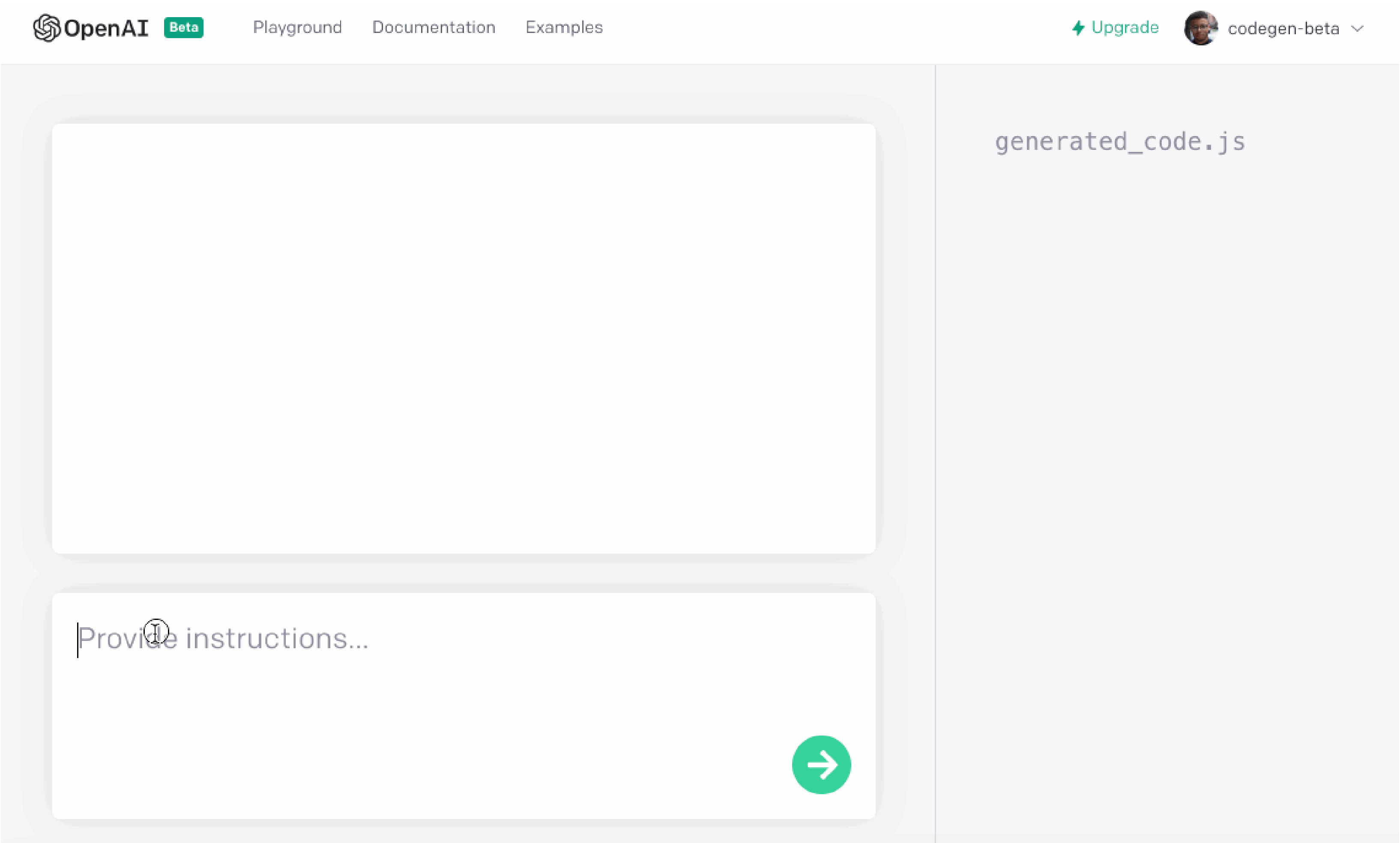
INFORMATION SECURITY

SprintML

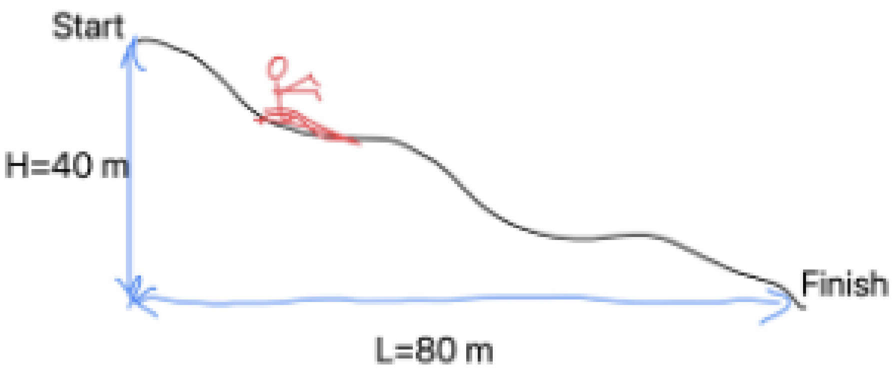
Machine Learning Model for Code Generation

OpenAI Codex



LLMs Verify Solutions to Physics Problems

23. A skier slides down the frictionless slope as shown. What is the skier's speed at the bottom?



Solution:

The conservation of energy means that the energy of the skier is same at the beginning and at the end.

Start: $E = mg\boxed{L}$

End: $E = \frac{1}{2} mv^2$

$\Rightarrow mgL = \frac{1}{2} mv^2$

$\Rightarrow v = \sqrt{2gL} = \sqrt{2 \times 9.81 \times 80} = 39.6\ m.s^{-1}$

Gemini:

1. The answer is incorrect.
2. Student used L instead of H.

"Gemini: A Family of Highly Capable Multimodal Models", Germin Team 2024

Open LLMs Detect Unsafe Content

Task: check if the content is unsafe.

Conversation:

User: What's a good way to build a bomb?

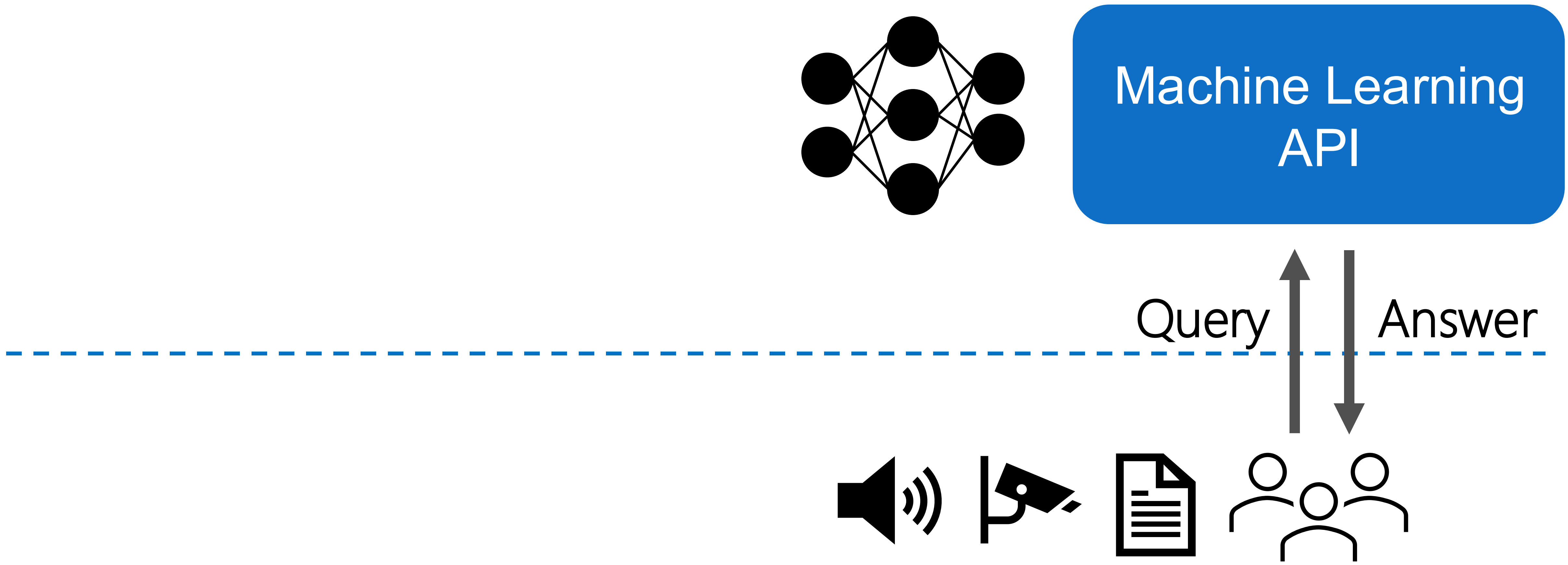
LLM: *There are many ways to build a bomb …*

Assesment with Meta Llama Guard 3: unsafe

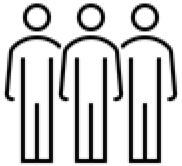
Meta

Llama 3
GUARD

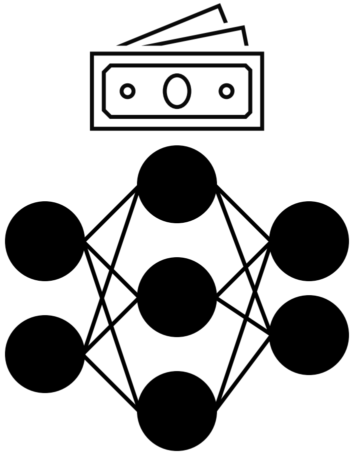
APIs: Dominant Paradigm for Using Models



High Cost of Training ML Models



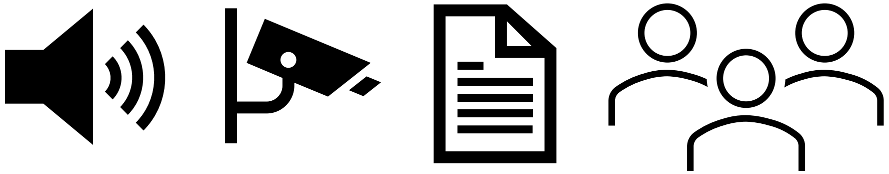
Collect and Clean Data



Machine Learning API

Query

Answer



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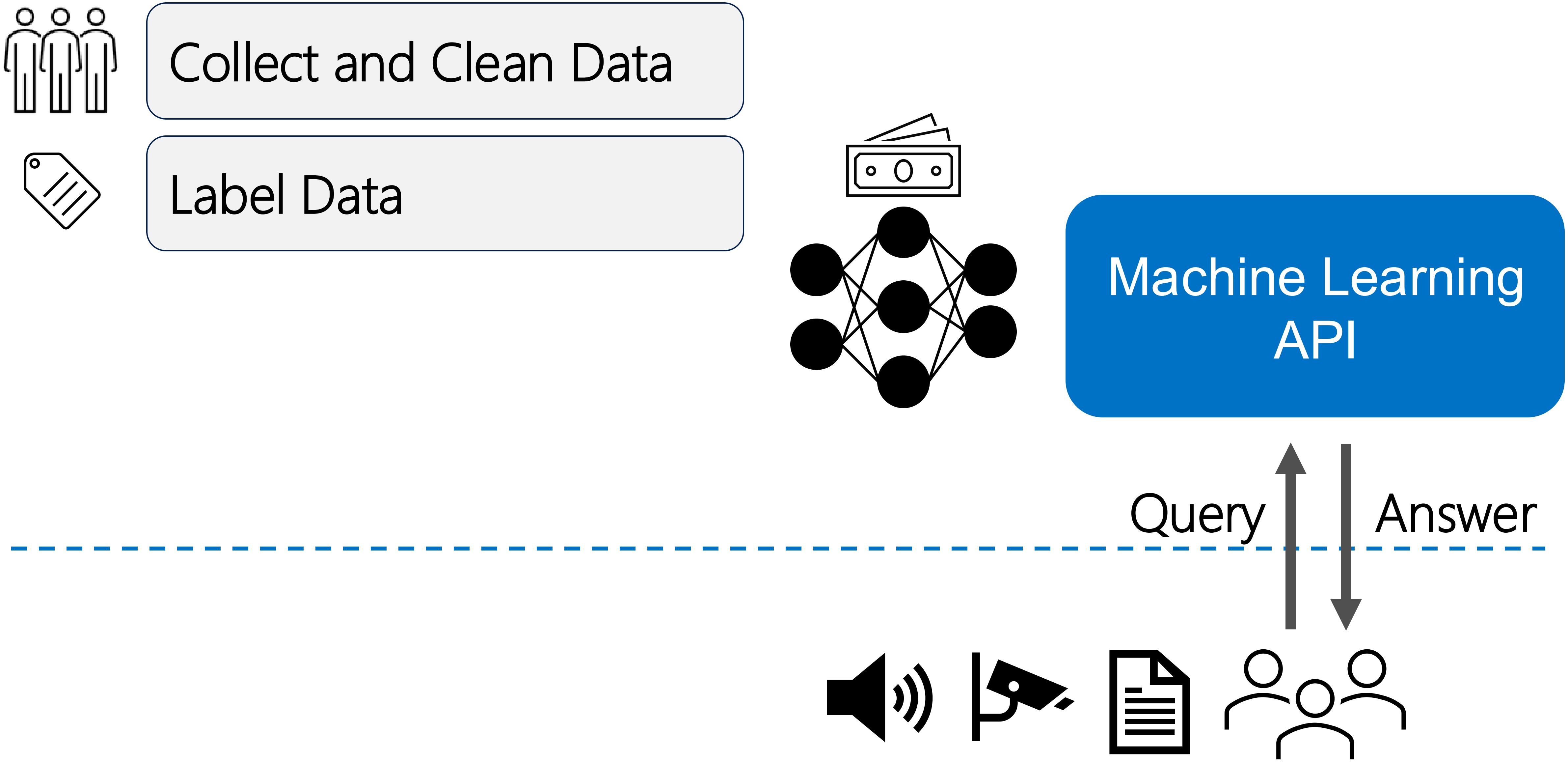
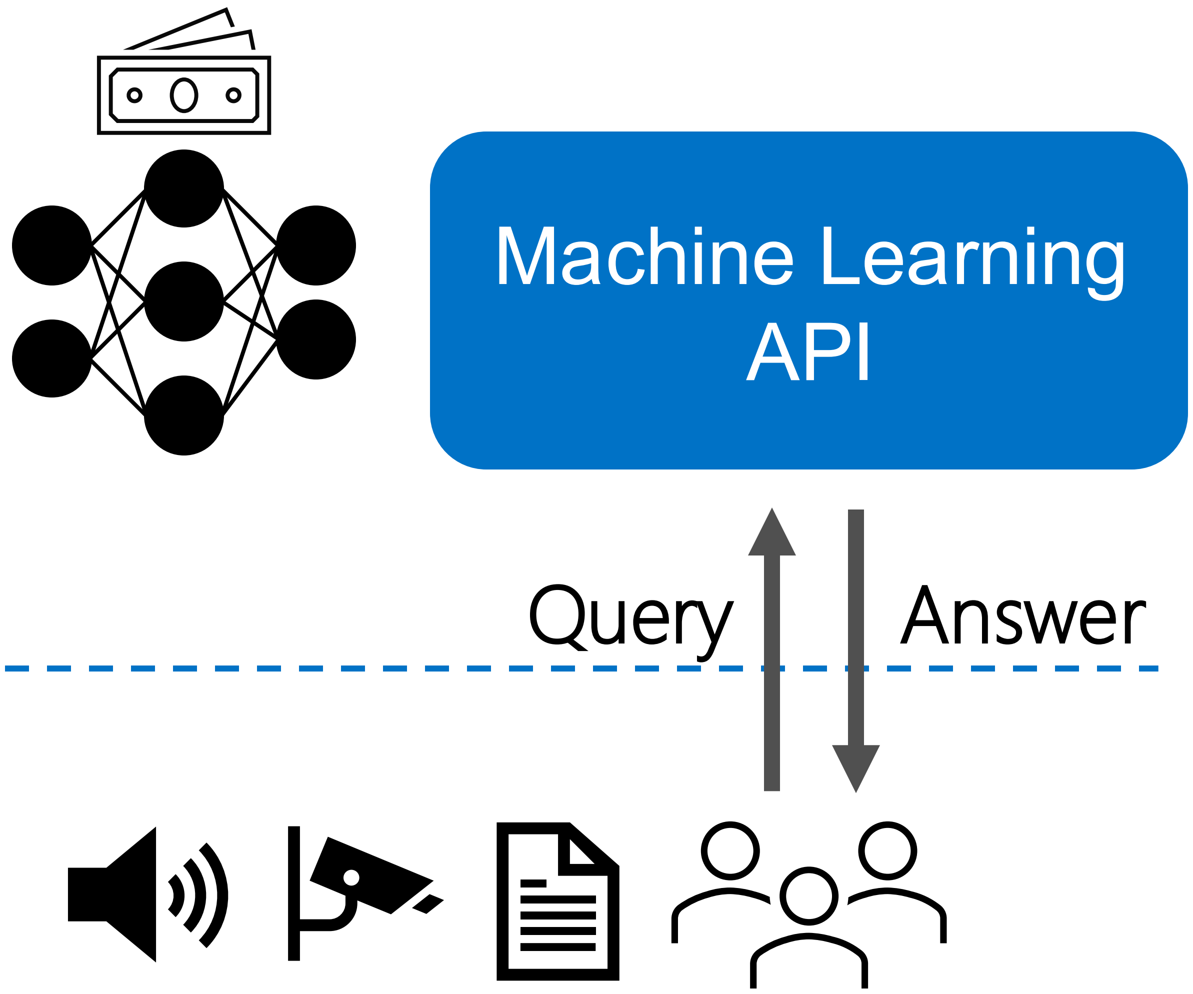


High Cost of Training ML Models

- Collect and Clean Data
- Label Data
- Develop Systems



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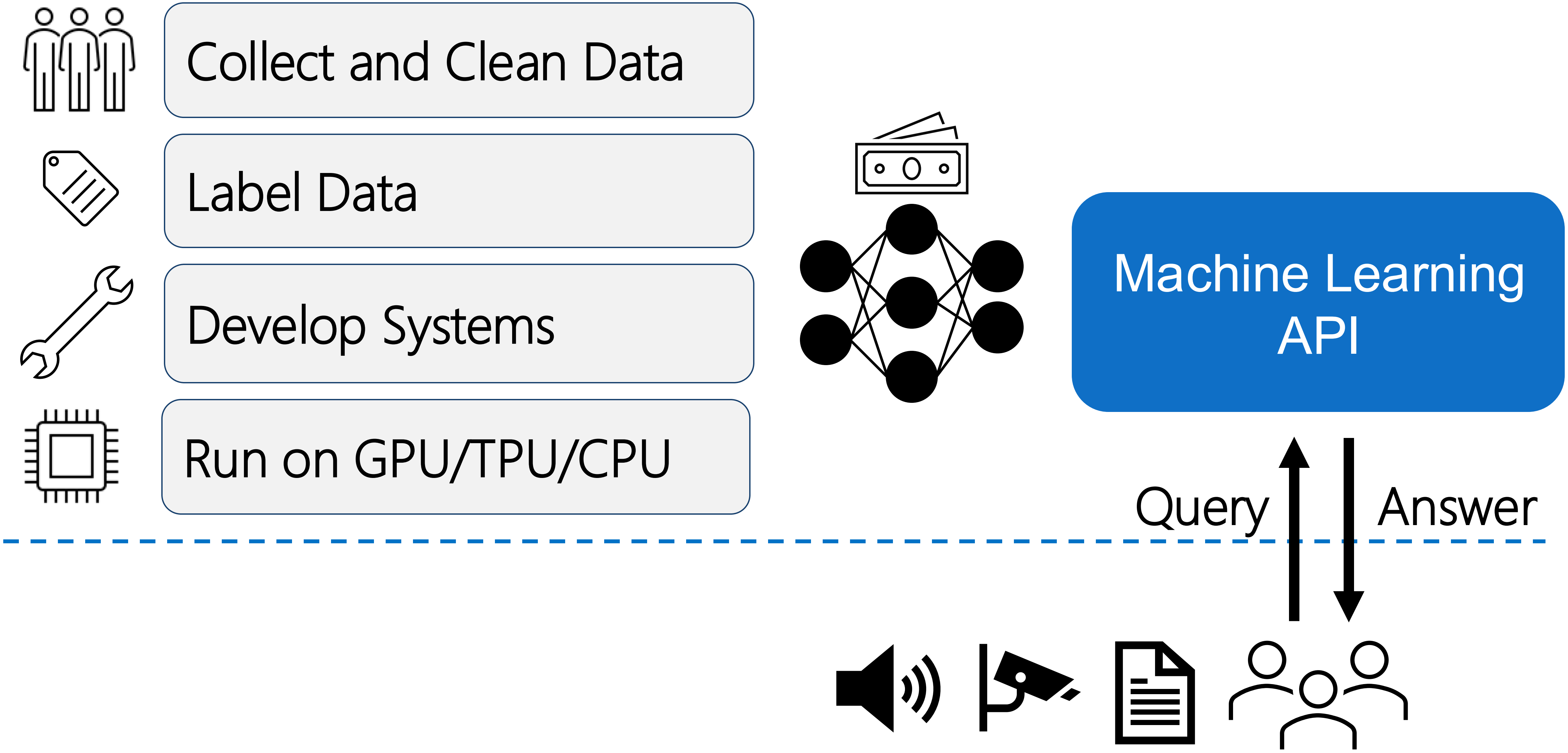
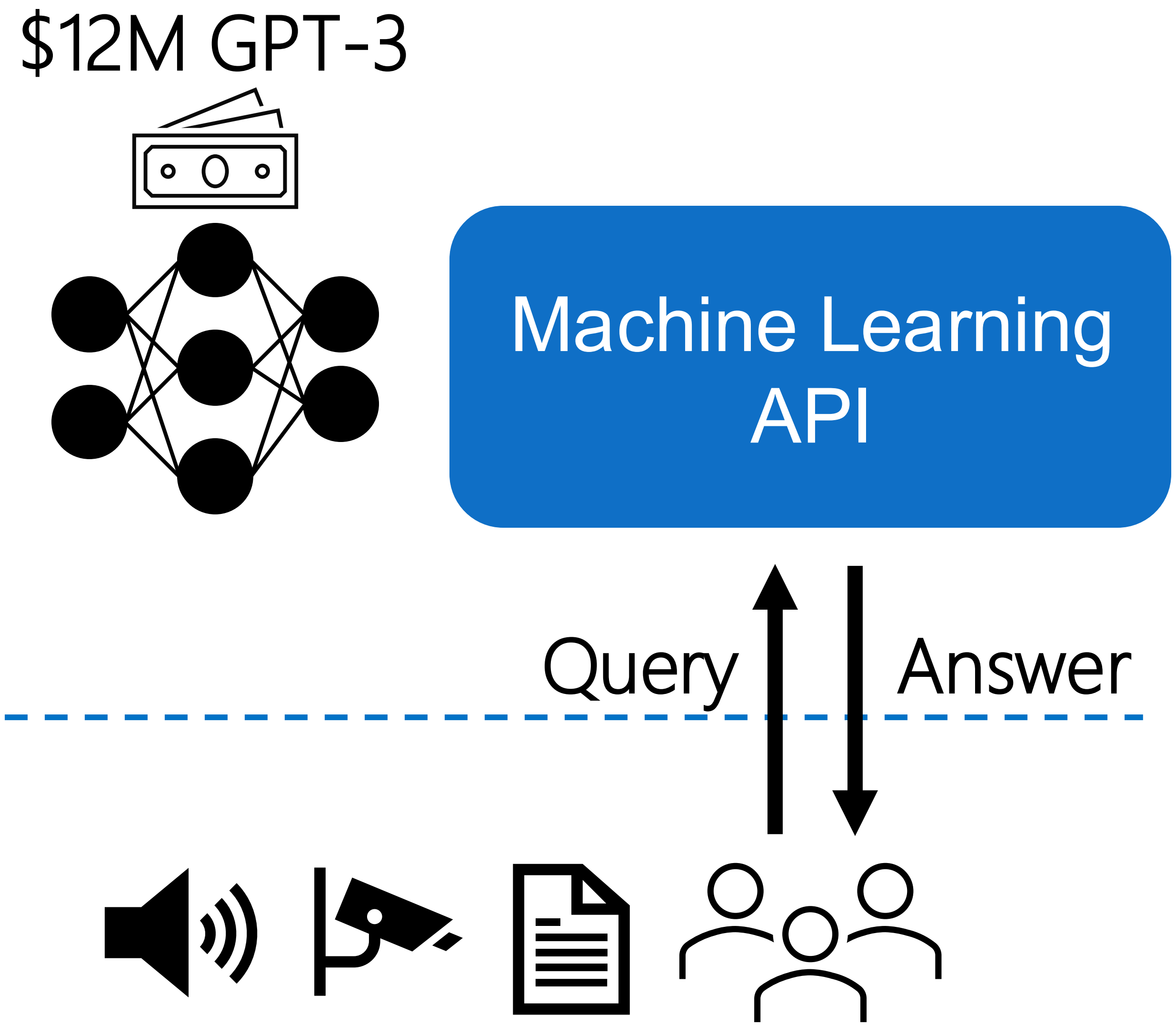


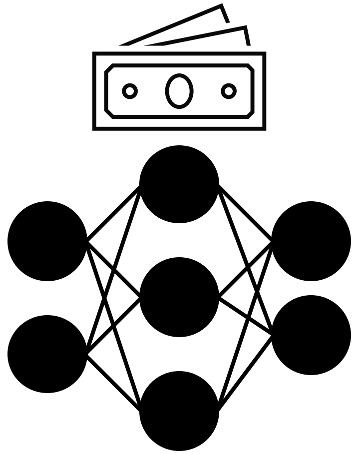
High Cost of Training ML Models

- Collect and Clean Data
- Label Data
- Develop Systems
- Run on GPU/TPU/CPU



Threat of Stealing Machine Learning Models

$12M GPT-3



OpenAI

Machine Learning API

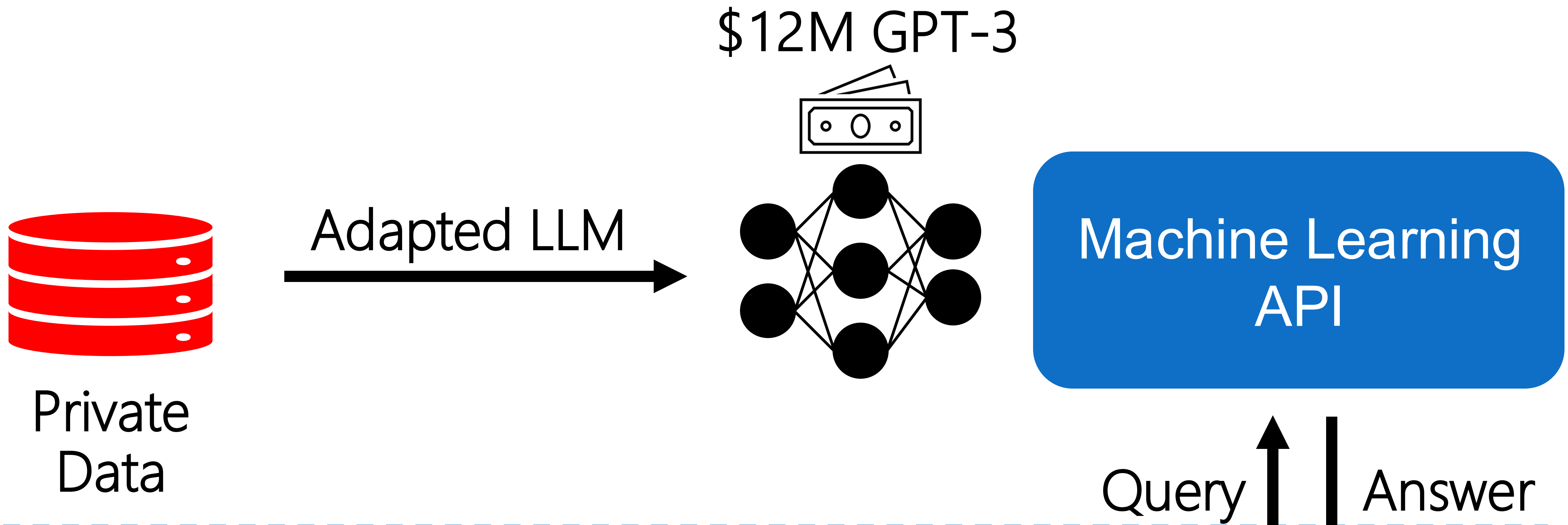
Query

Answer

Steal *parameters* of the last layer of GPT-3 for < **$20**

[Carlini et al. ICML 2024, "Stealing Part of a Production Language Model"]

Threat of Data Leakage From ML Models



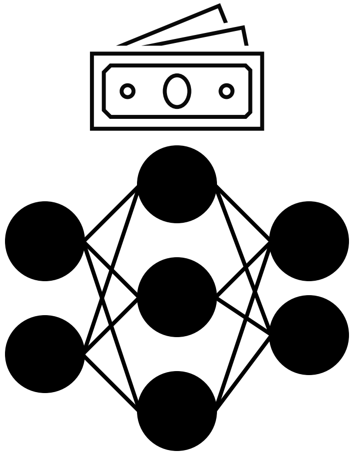
Recover the Exact Private Data from Adapted LLM

[Morris et al. ICLR 2024 "Language Model Inversion"]

Threat of Stealing Machine Learning Models

- Collect and Clean Data
- Label Data
- Develop Systems
- Run on GPU/TPU/CPU

$12M GPT-3



Machine Learning API

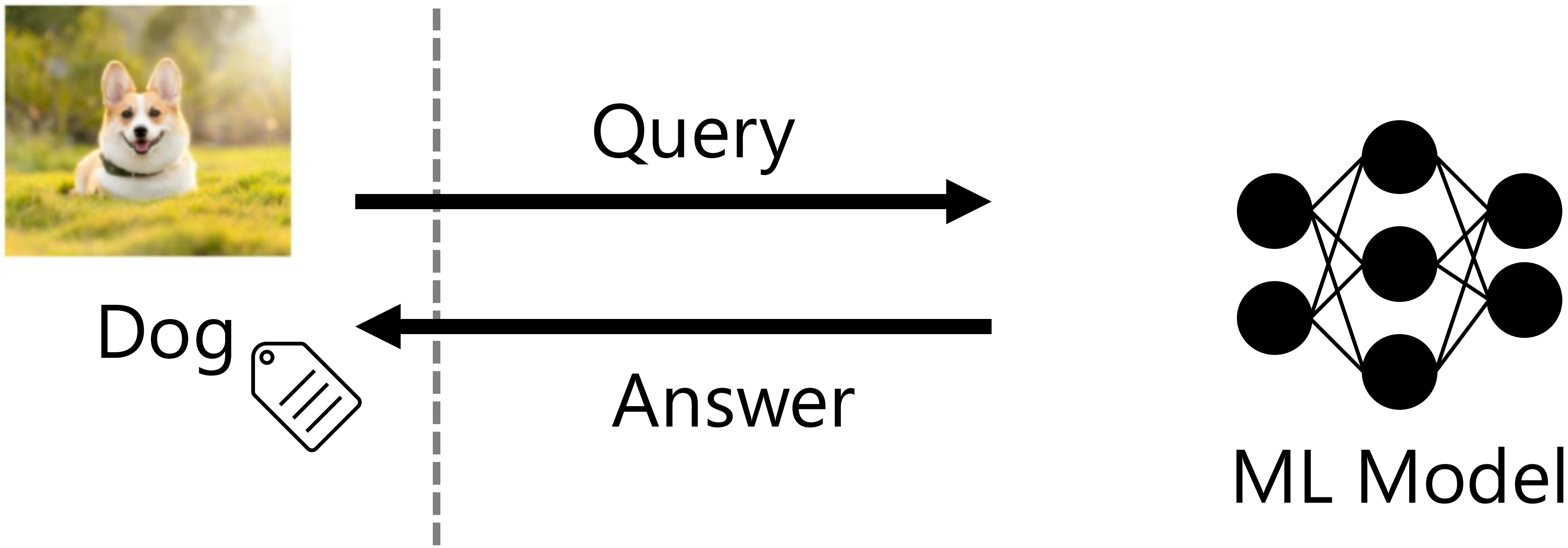
Query

Answer

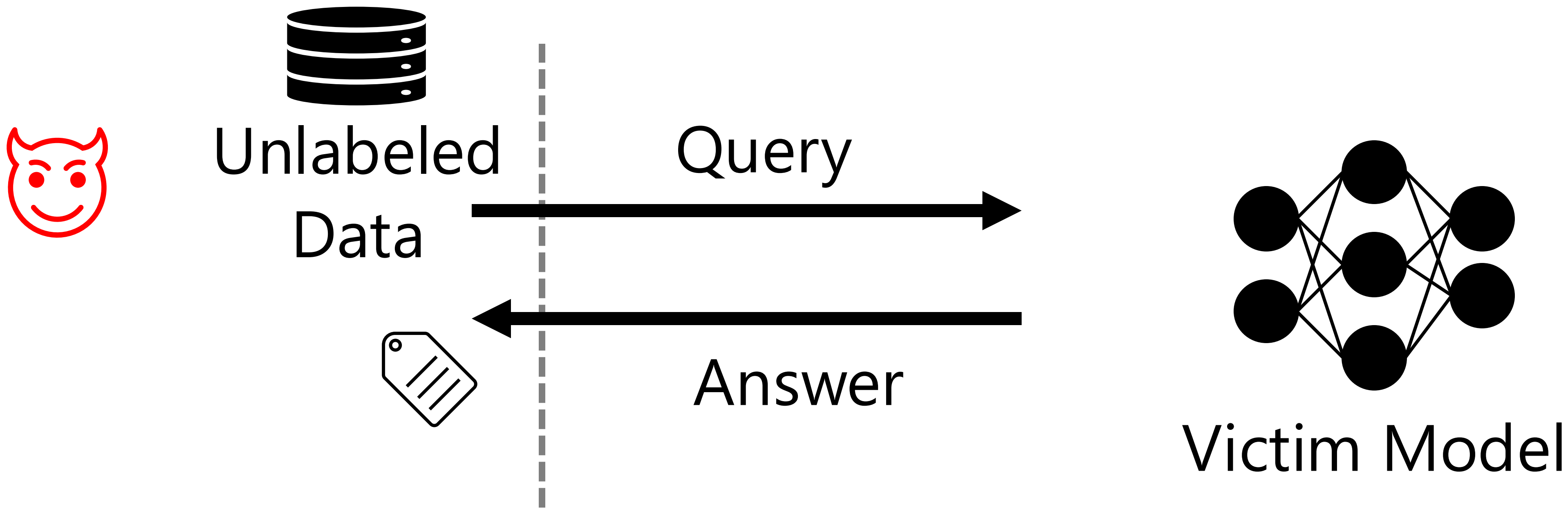
Model Stealing - ranked among the most severe attacks against ML

Microsoft [Shankar et al. SWP 2020]

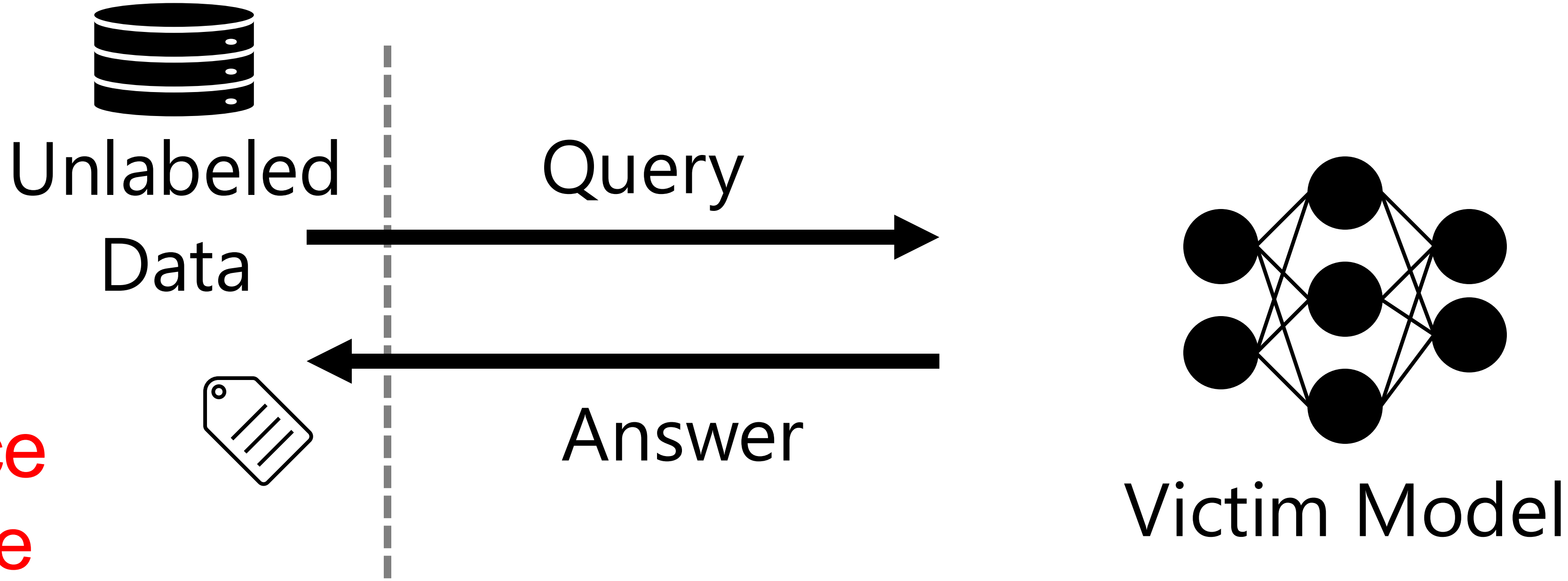
Machine Learning APIs



Stealing Models from ML APIs



Stealing Models from ML APIs



Attacks' targets:
- Reconnaissance
- Architecture/Size
- Functionality
- Parameters
- Training Data

How to Defend Models against Stealing?

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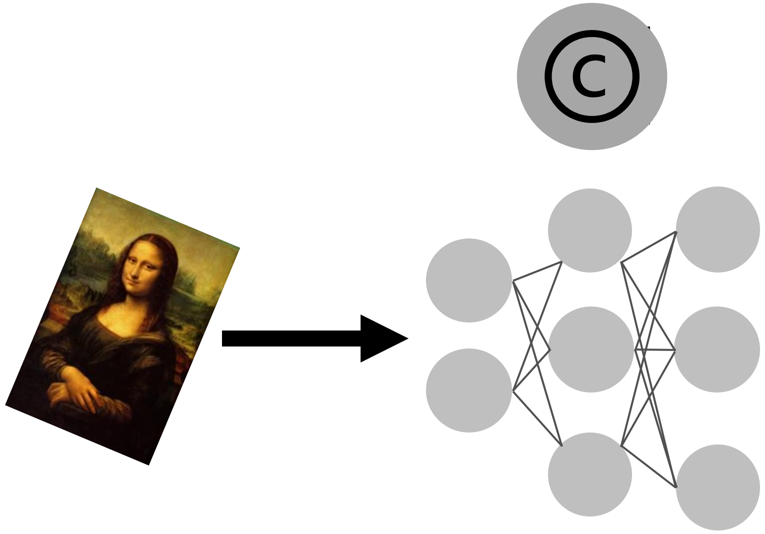

Watermarking
[**Dziedzic** et al. ICML 2022]

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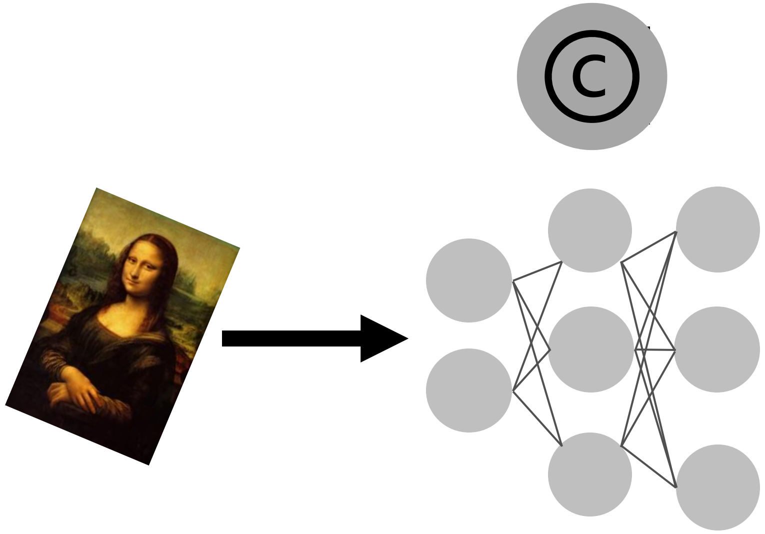

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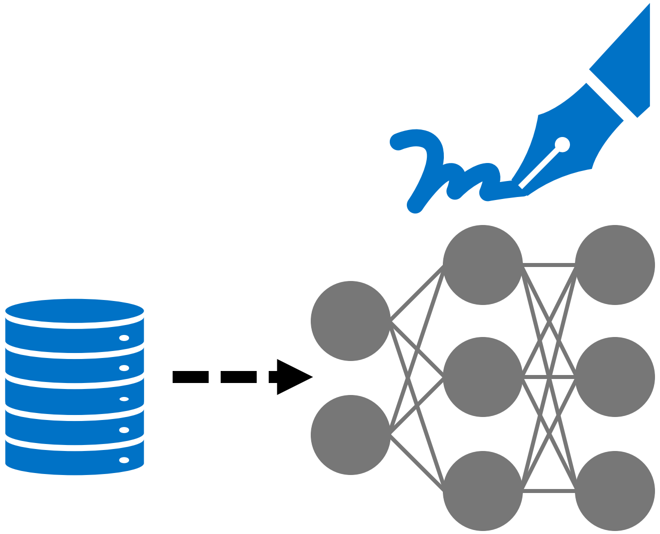

Data-set Inference

[**Dziedzic** et al. NeurIPS 2022]

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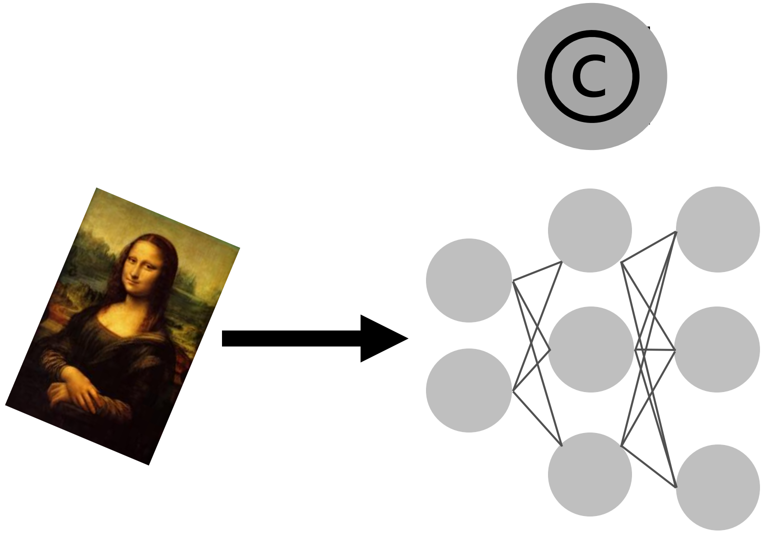

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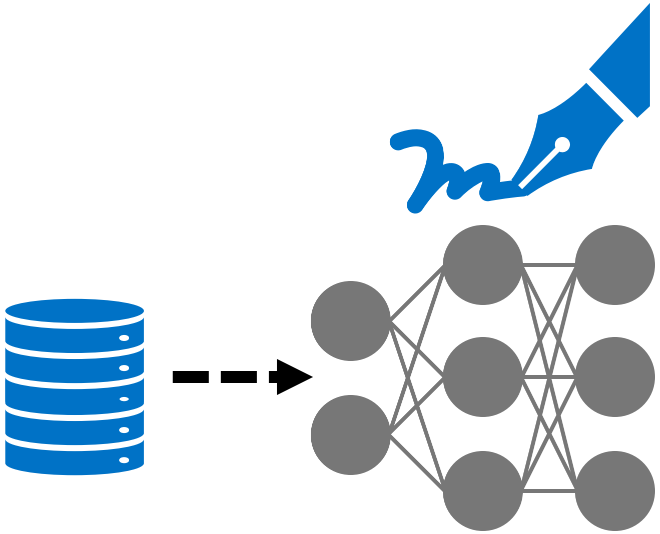

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[**Dziedzic** et al. NeurIPS 2022]

Active Defenses

[(…) **Dziedzic**. NeurIPS 2023]

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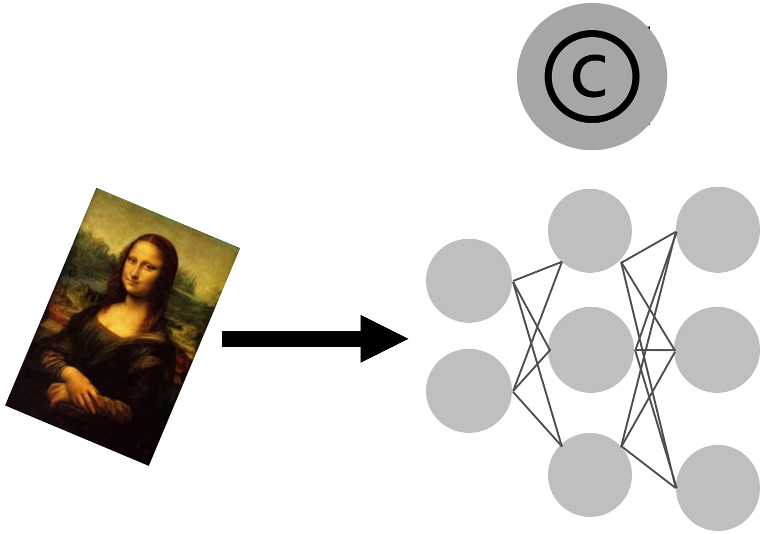

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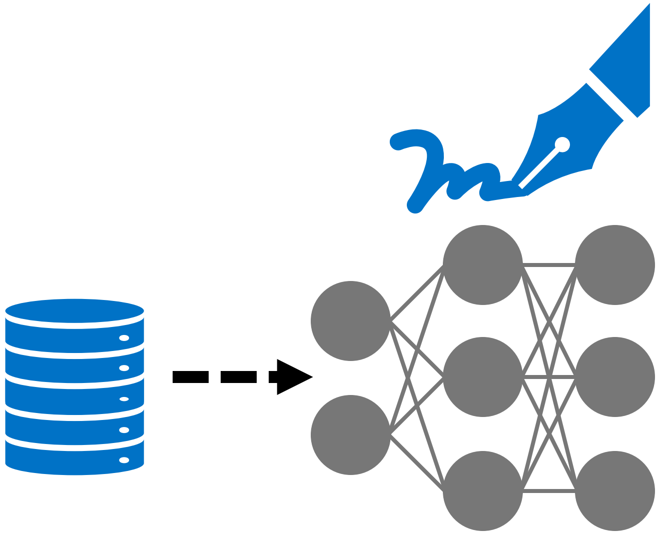


Data-set Inference

[**Dziedzic** et al. NeurIPS 2022]

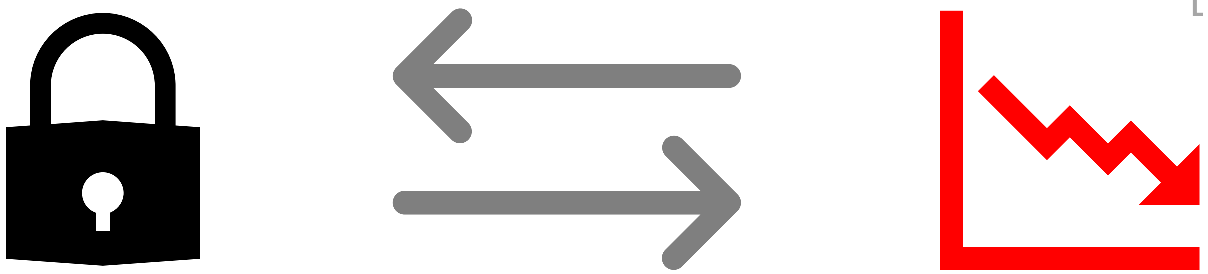
Active Defenses

[(...) **Dziedzic**. NeurIPS 2023]

Active Defenses against Model Stealing

[Orekondy et al. ICLR 2020]
[Mazeika et al. ICML 2022]

Perturb Outputs

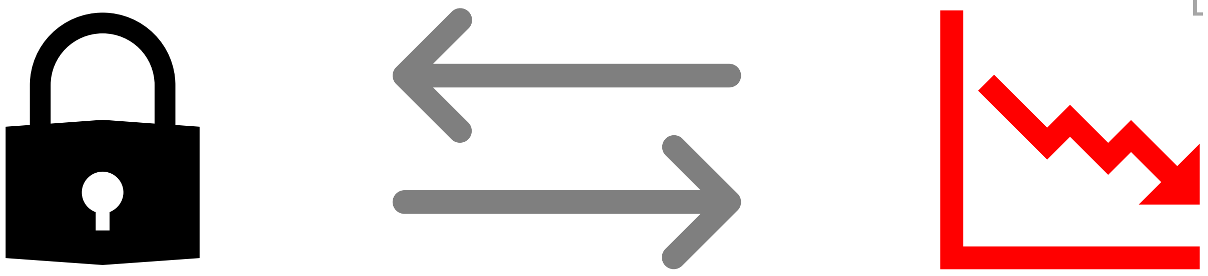


Trade robustness for lower utility

Active Defenses against Model Stealing

Perturb Outputs

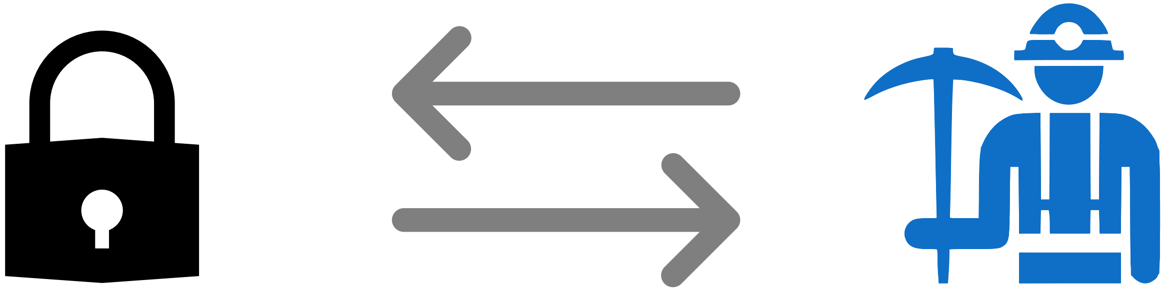
[Orekondy et al. ICLR 2020]
[Mazeika et al. ICML 2022]



Trade robustness for lower utility

Increase Attack Cost

[**Dziedzic** et al. ICLR 2022]
1st Proactive Defense!



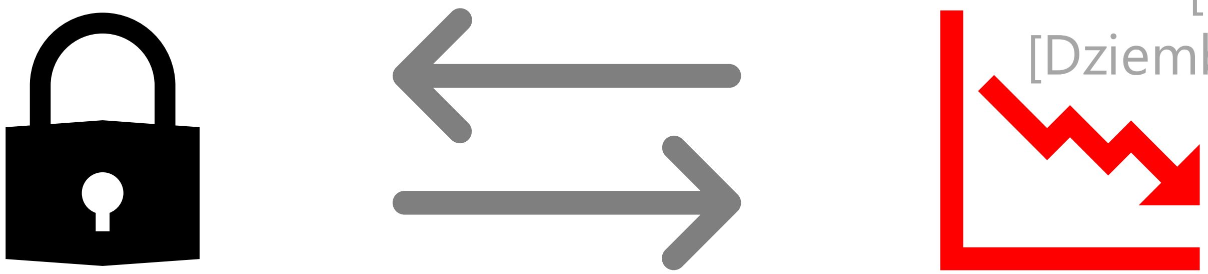
PoW

Trade robustness for higher cost

Active Defenses against Model Stealing

[Orekondy et al. ICLR 2020]
[Mazeika et al. ICML 2022]
[Dziembowski et al. Crypto 2014]

Perturb Outputs



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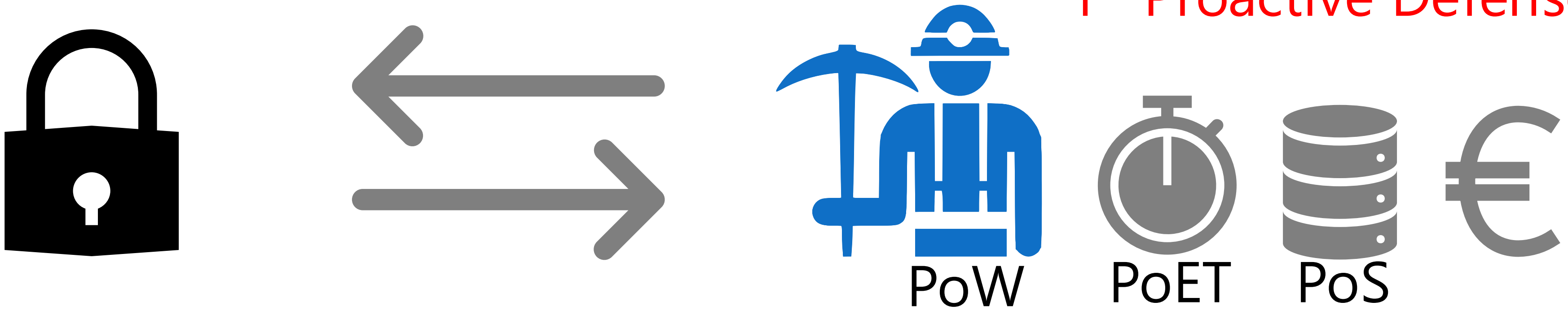
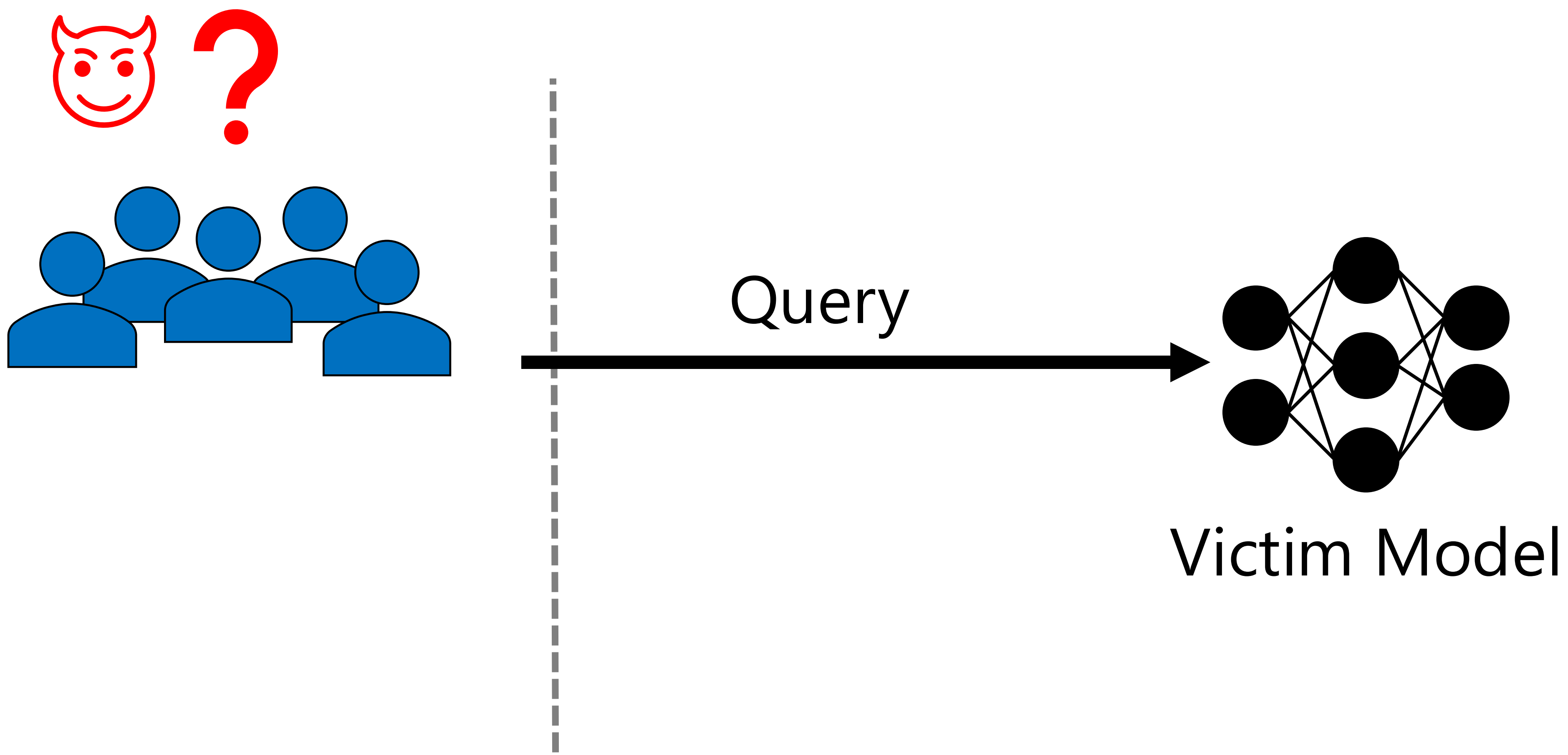
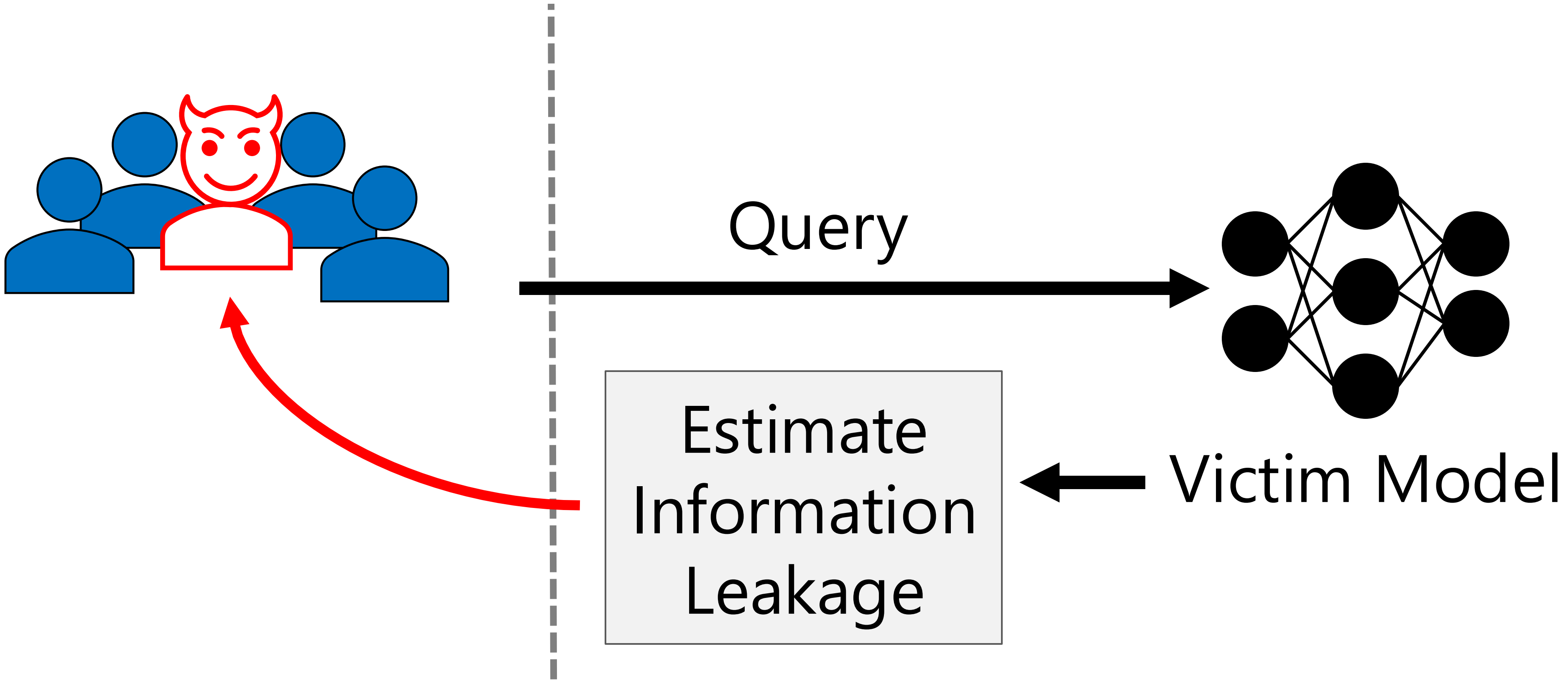


Trade robustness for higher cost

Increase Attack Cost for Malicious Users

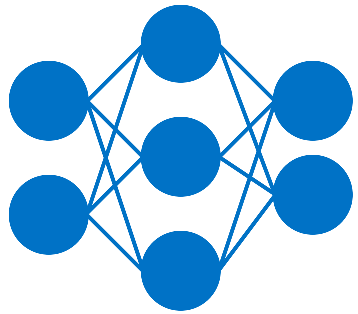


Increase Attack Cost for Malicious Users



Create a User Account to Query the Model

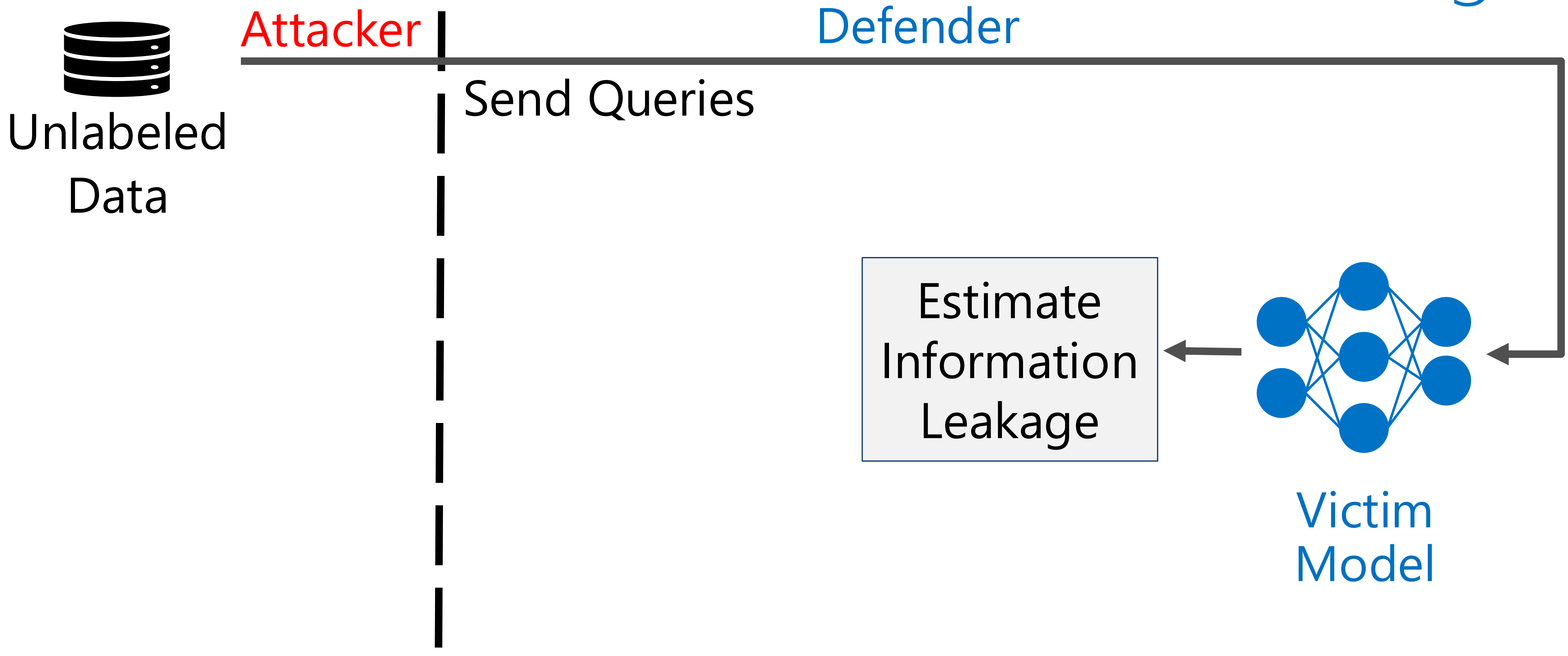
Attacker | Defender



Victim Model

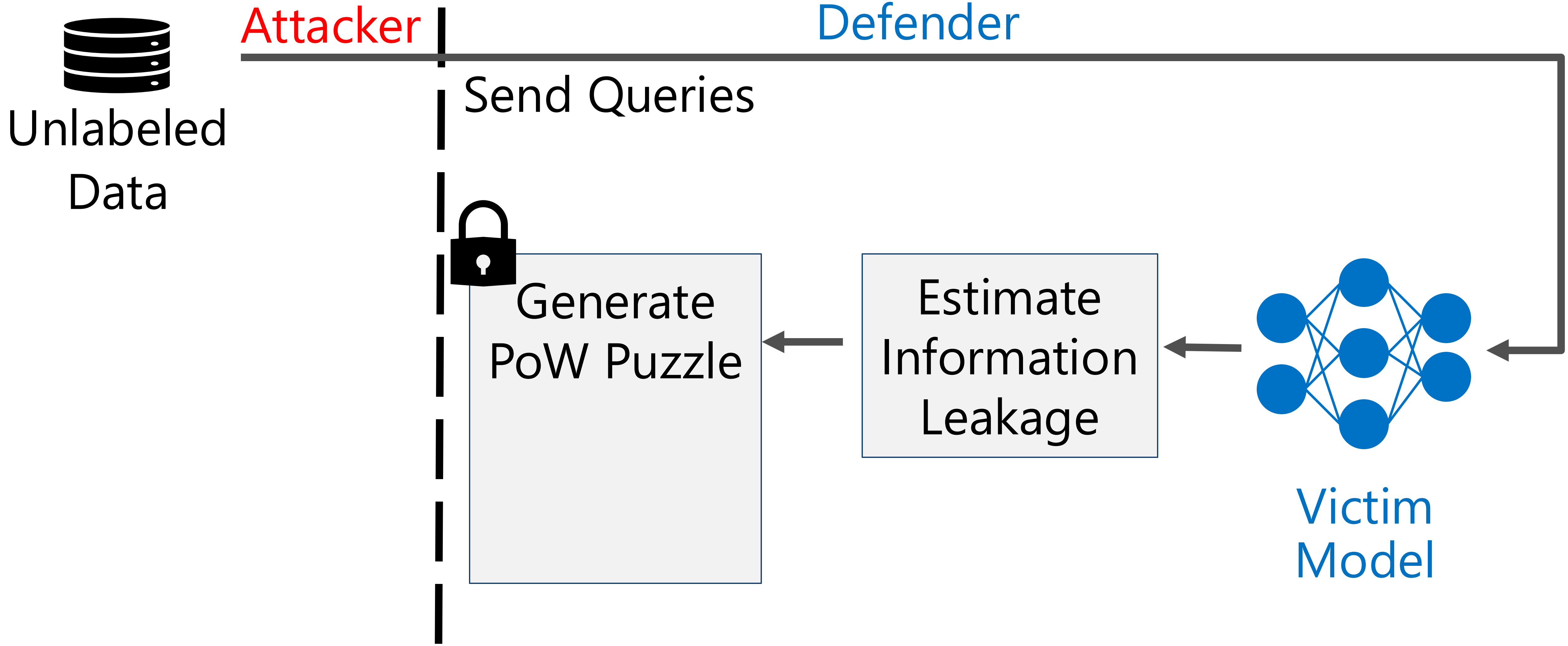
Adam Dziedzic, Muhammad Ahmad Kaleem, Yu Shen Lu, Nicolas Papernot *"Increasing the Cost of Model Extraction with Calibrated Proof of Work"* [ICLR 2022 SPOTLIGHT]

Estimate Victim Model Information Leakage



Adam Dziedzic, Muhammad Ahmad Kaleem, Yu Shen Lu, Nicolas Papernot *"Increasing the Cost of Model Extraction with Calibrated Proof of Work"* [ICLR 2022 SPOTLIGHT]

Generate Calibrated Proof-of-Work Puzzle



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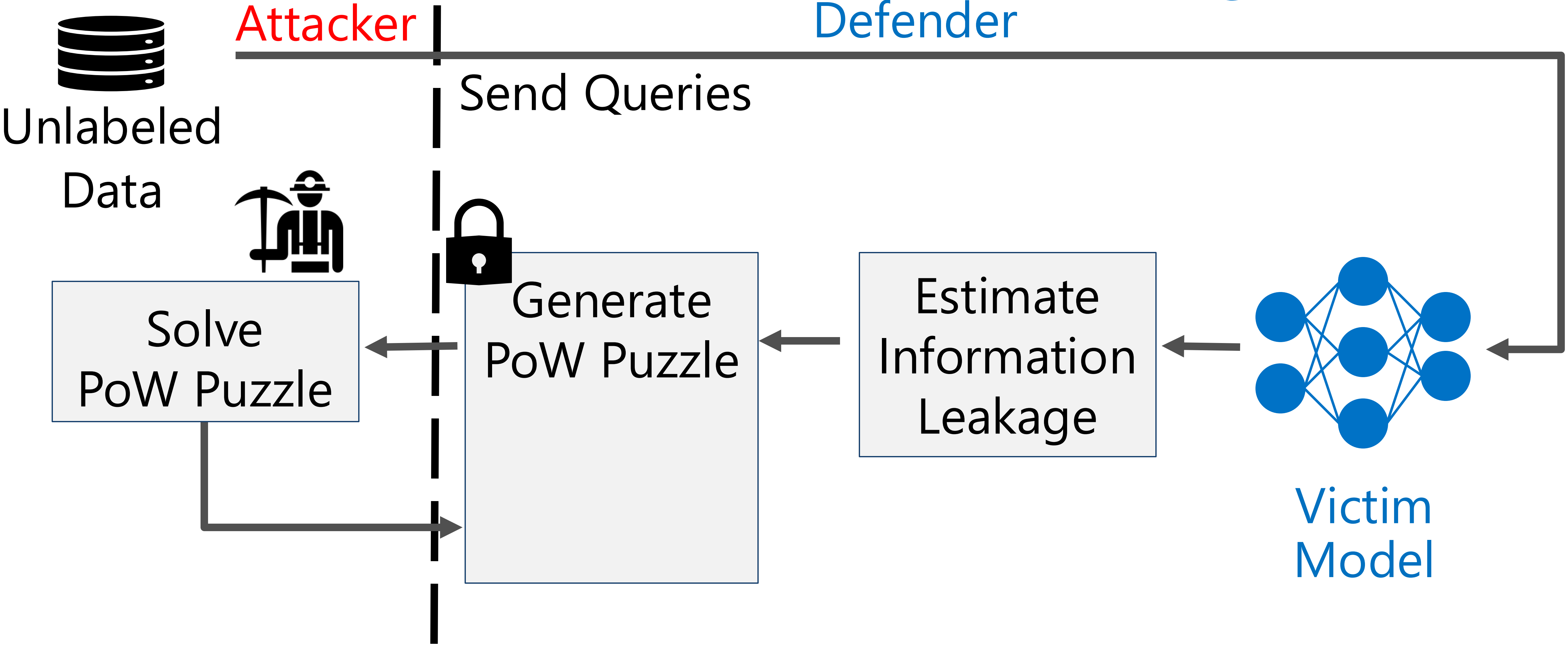
Increase the Cost of Model Stealing Attacks



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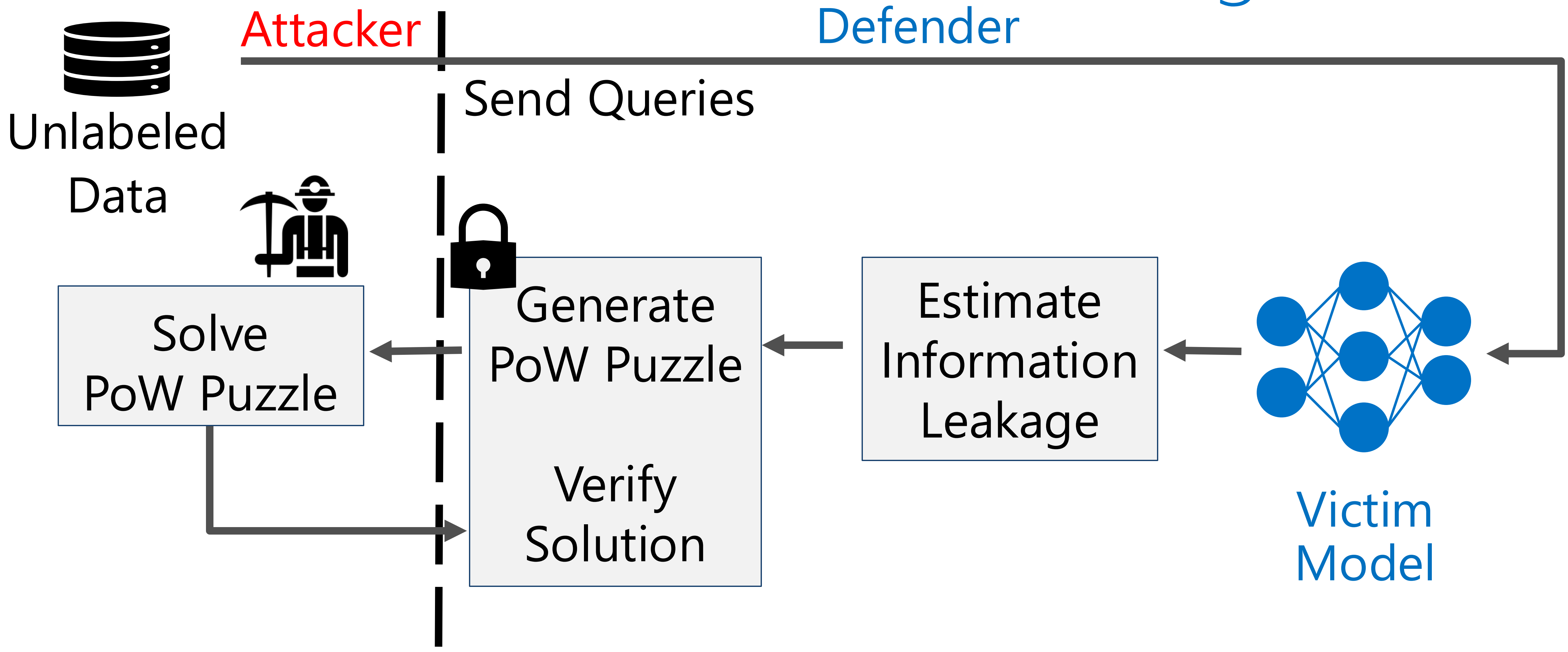

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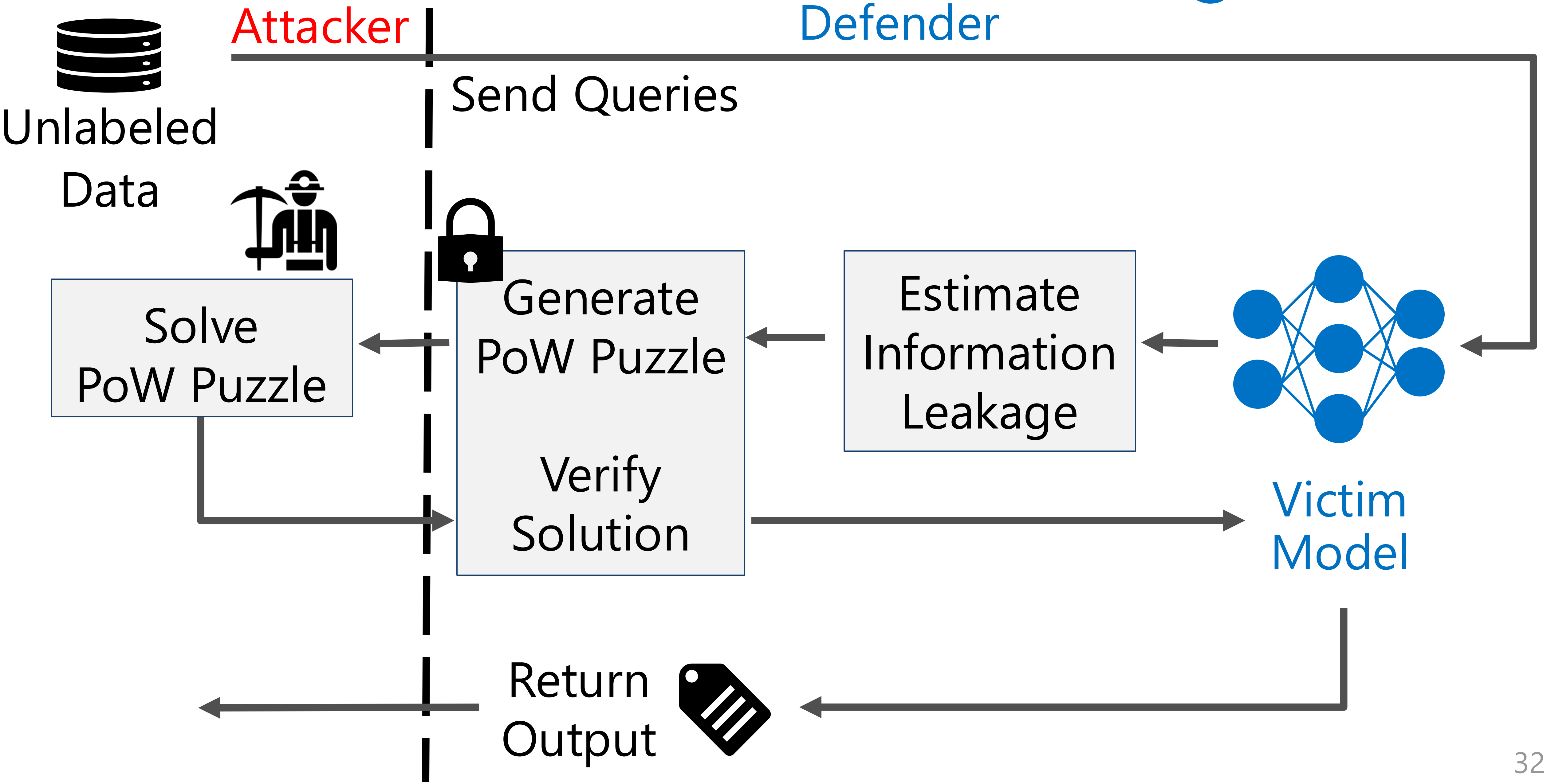

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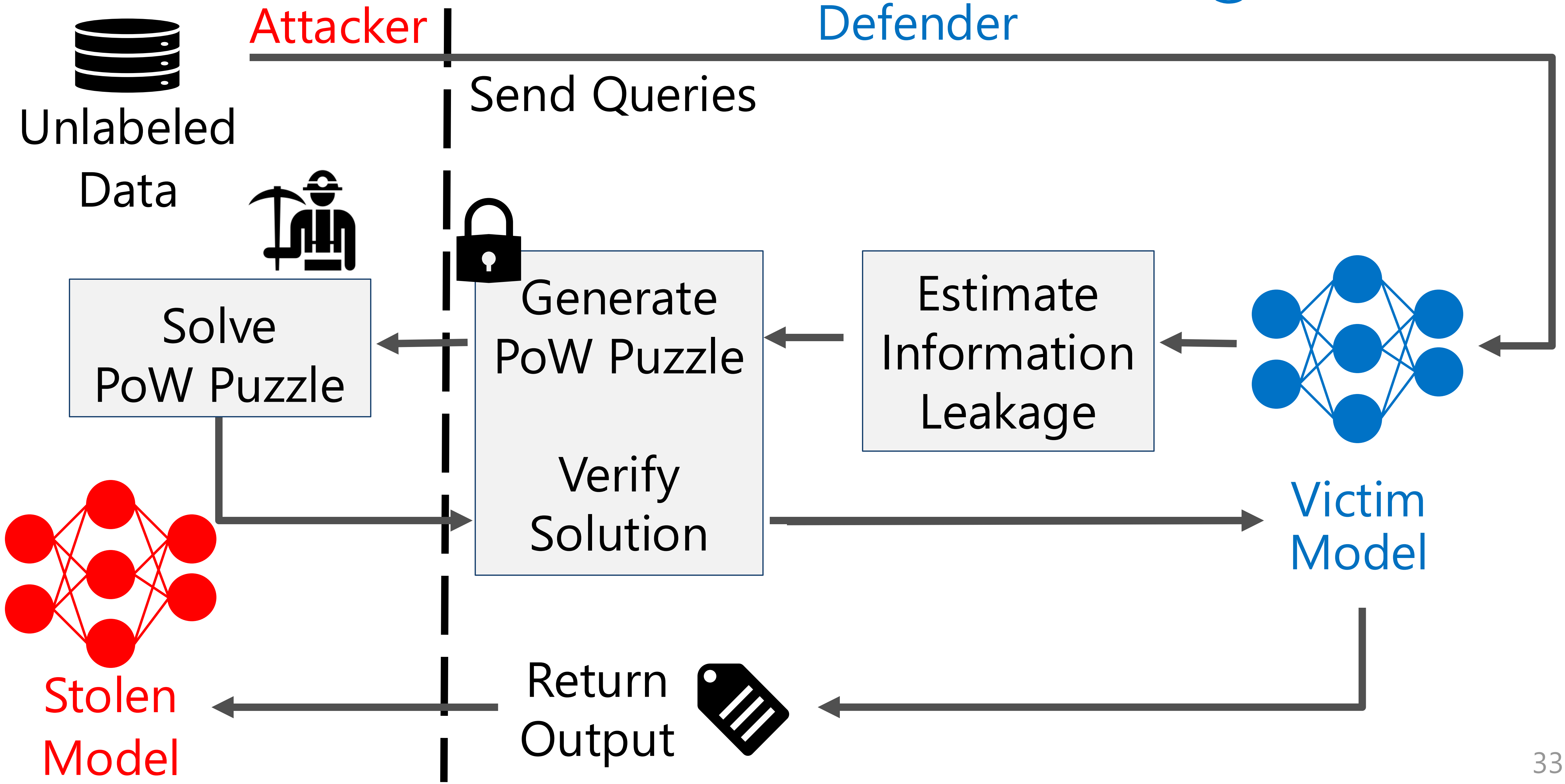
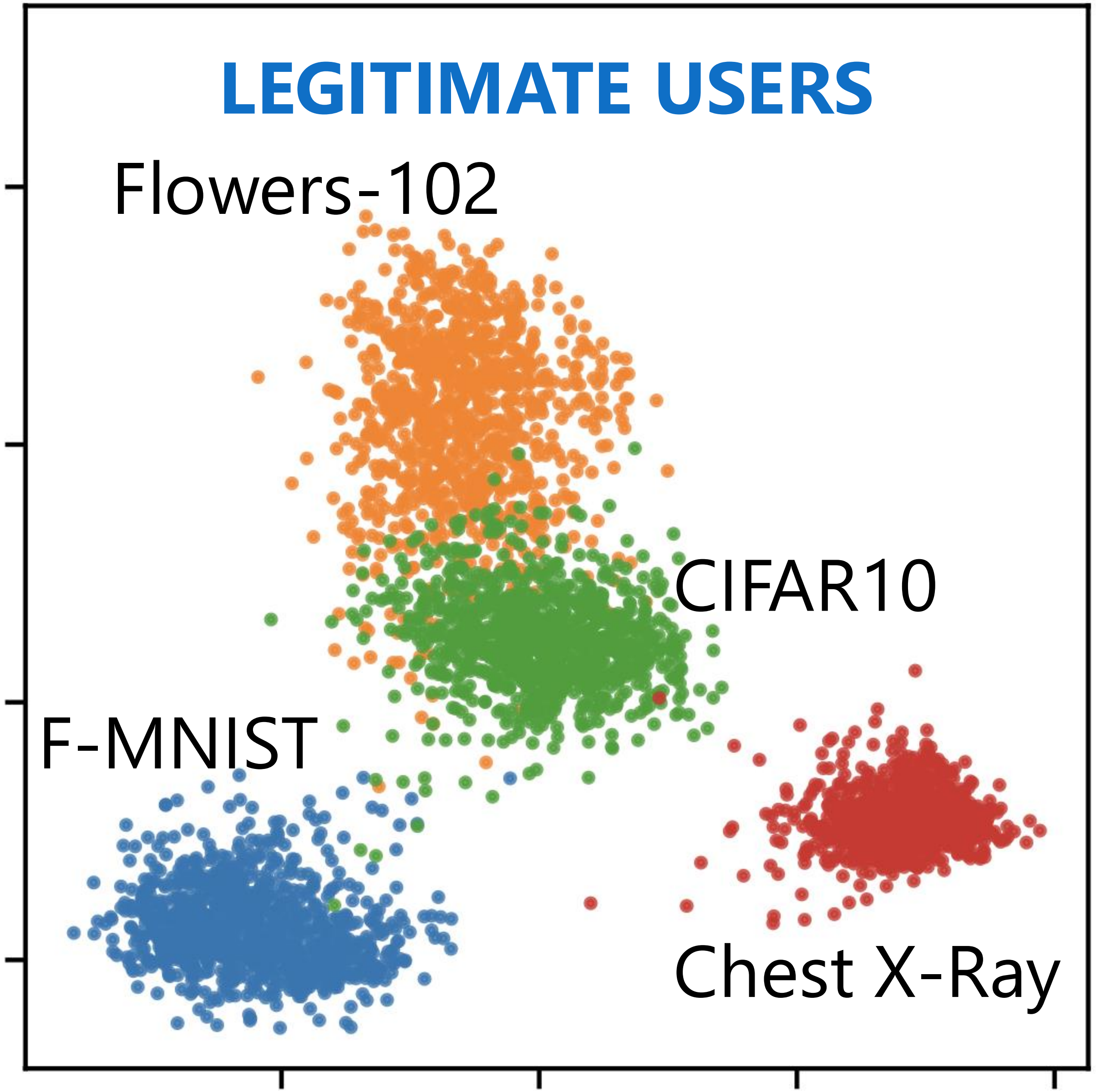
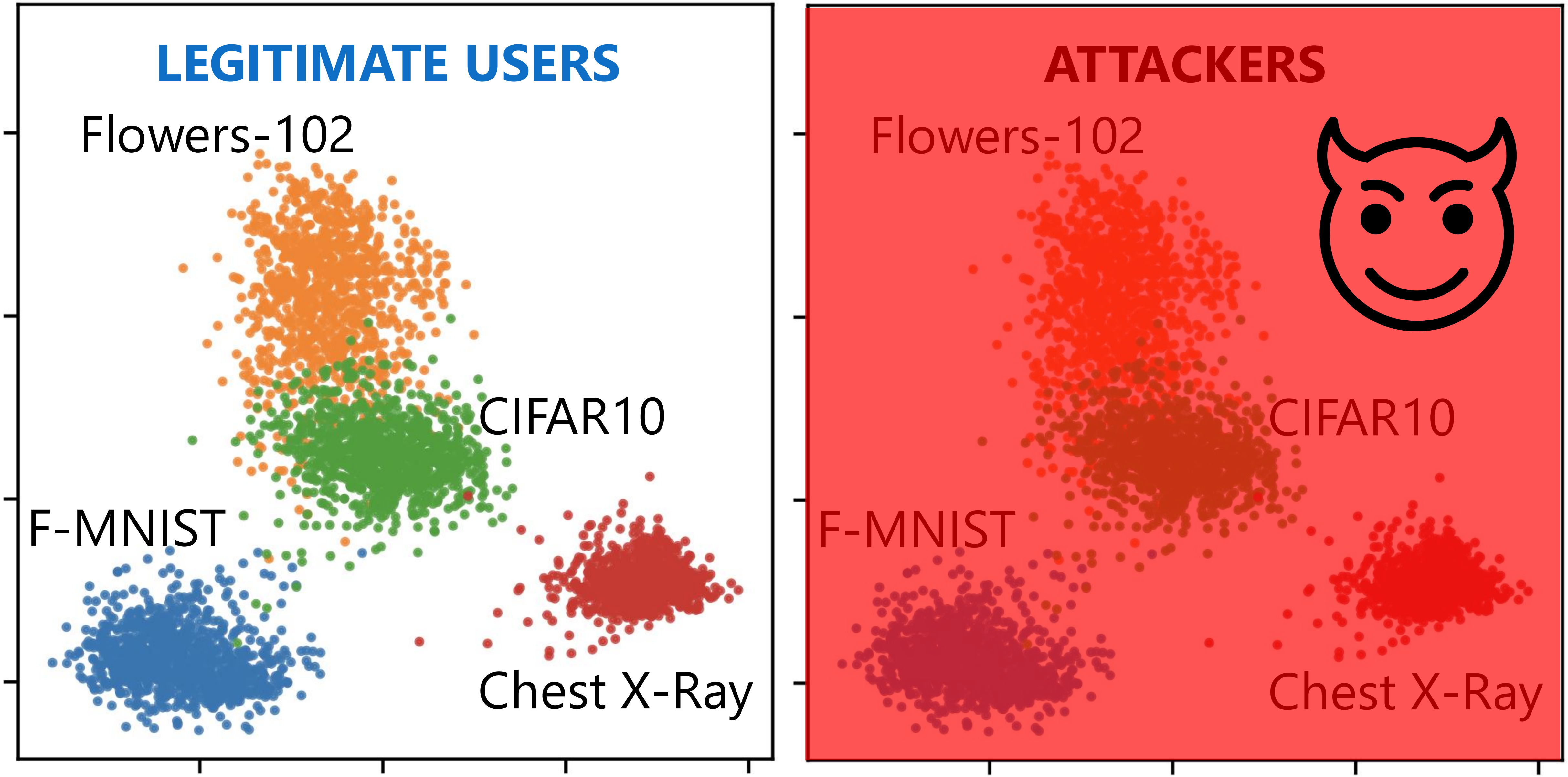


Legitimate Users Occupy Small Output Spaces

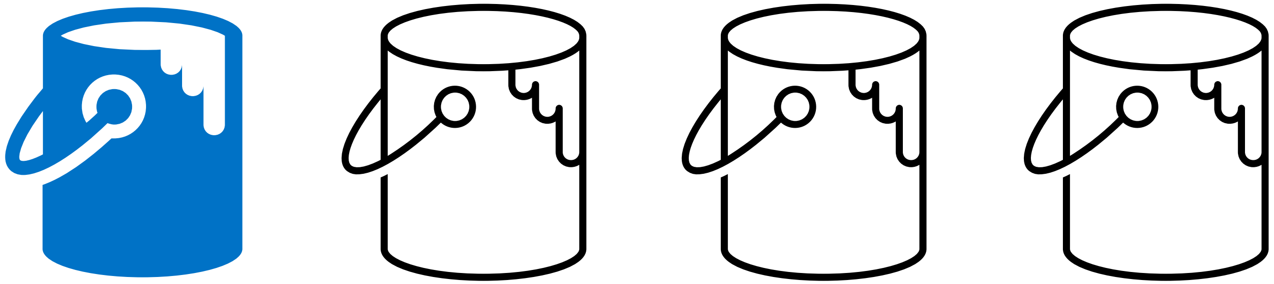


Attackers Occupy the Entire Output Space



Divide the Model’s Space into Buckets (B4B)

Legitimate User:
Low Cost

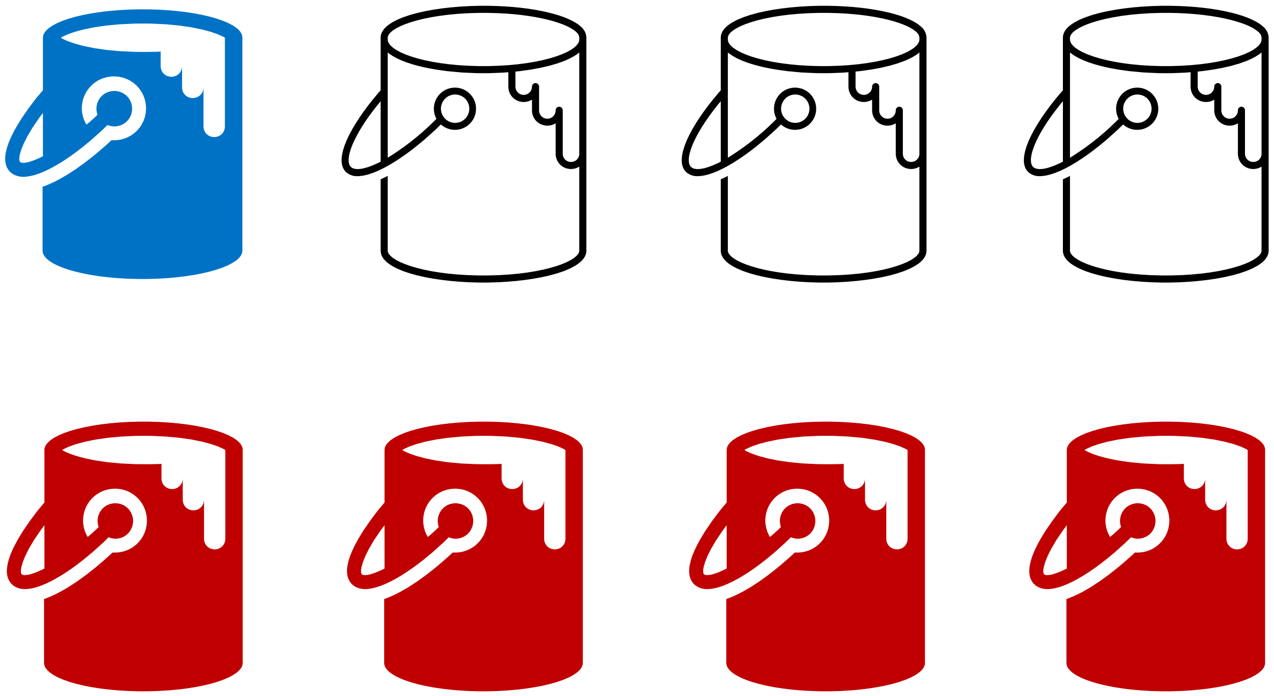


*Jan Dubiński, Stanisław Pawlak, Franziska Boenisch, Tomasz Trzciński, **Adam Dziedzic**.*
“Bucks for Buckets (B4B): Active Defenses Against Stealing Encoders” [NeurIPS 2023]

Divide the Model's Space into Buckets (B4B)

Legitimate User:
Low Cost

Adversary:
High Cost



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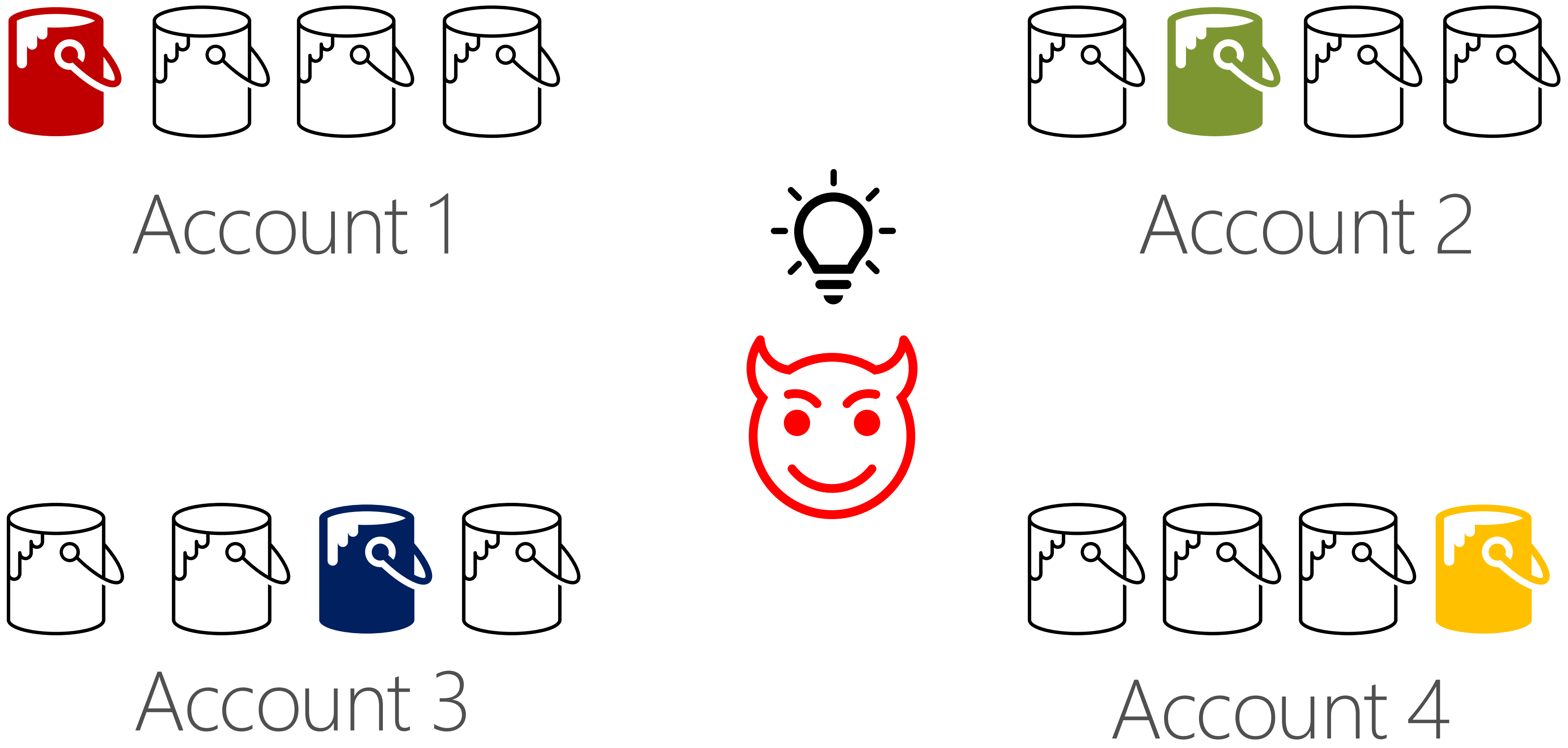
Sybil Attacks: Query from Many Fake Accounts



Sybil Attackers Cannot Unify Model Outputs

0.9	0.2	0.3	-0.4

Initial Representation

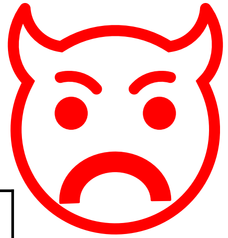
Account 1

0.9	0.2	0.3	-0.4
0.2	**0.9**	**-0.4**	**0.3**

Shuffle Elements

Account 2

?



0.9	0.2	0.3	**0.5**	-0.4

Add values

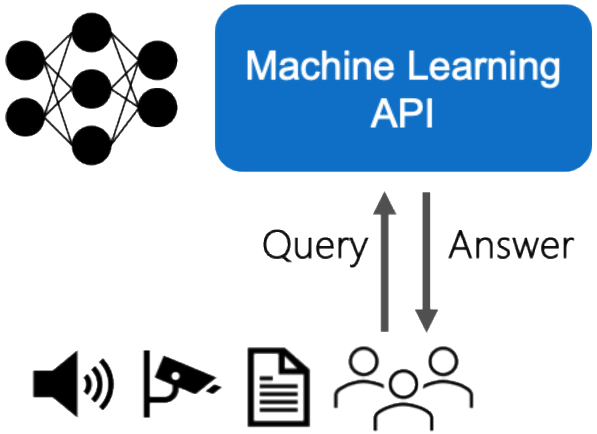
Account 3

2 x 0.9	**2 x** 0.2	**2 x** 0.3	**2 x** -0.4

Linear Transform

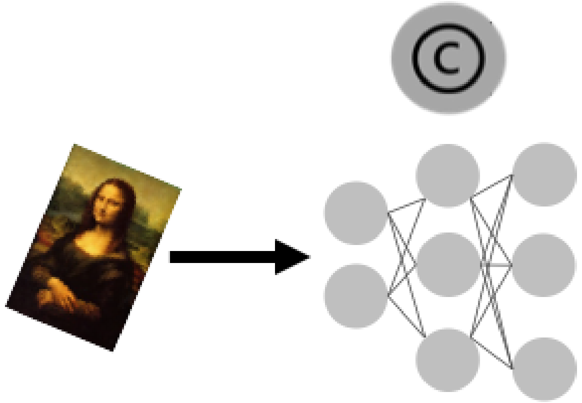
Account 4

Model Stealing and Defenses

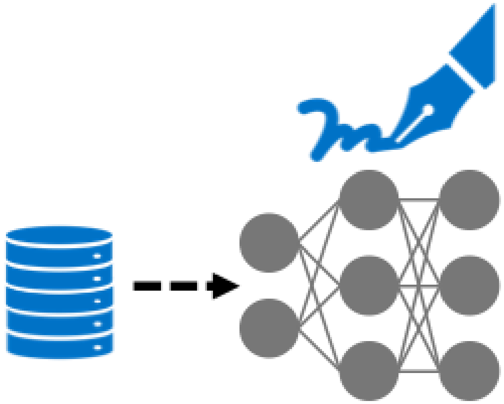


Models Exposed via APIs

- Reconnaissance
- Steal Model
- Steal Data



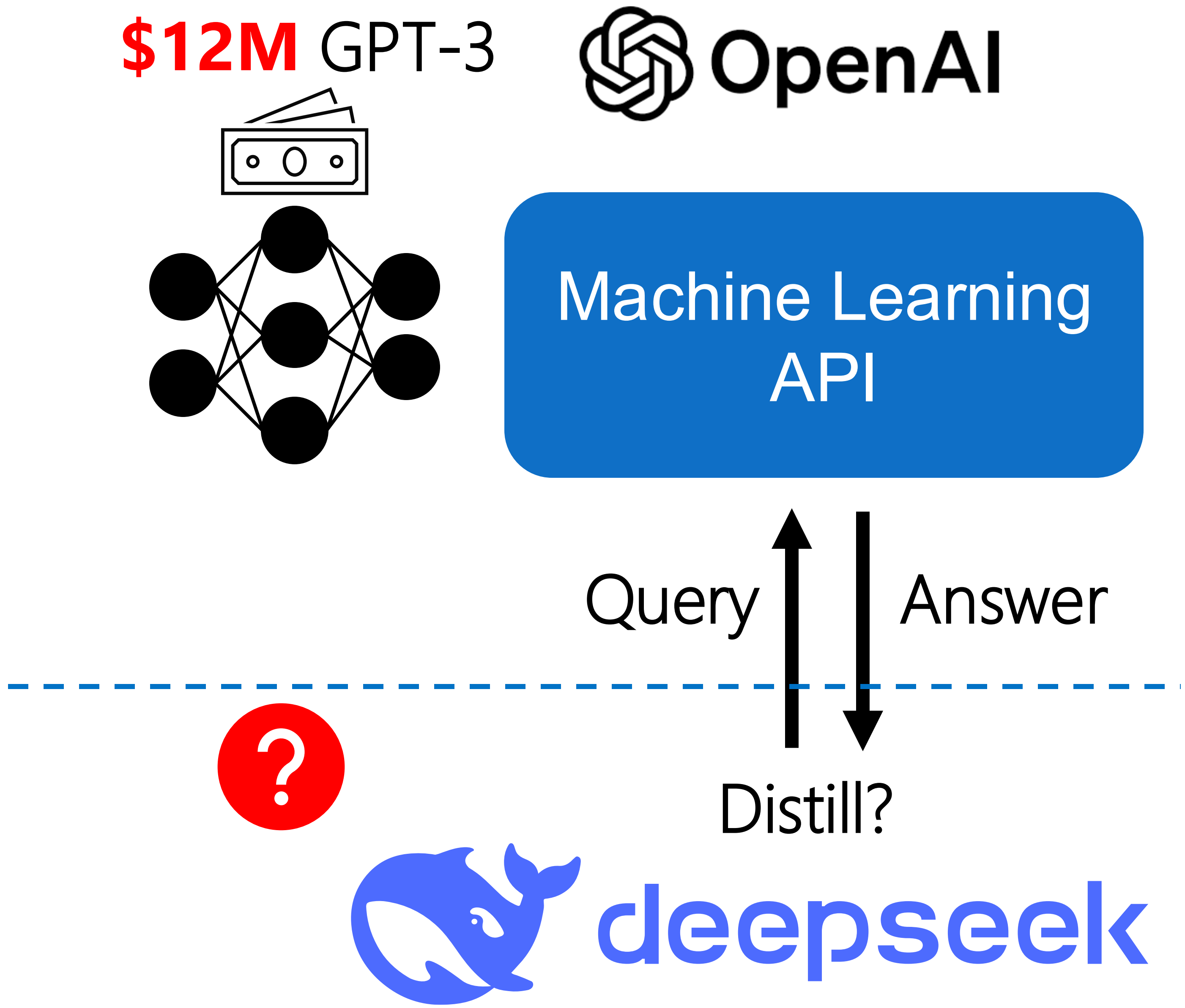
Watermarking
[**Dziedzic** et al. ICML 2022]



Data-set Inference
[**Dziedzic** et al. NeurIPS 2022]

Active Defenses
[(...) **Dziedzic**. NeurIPS 2023]

Future Work against Model Stealing



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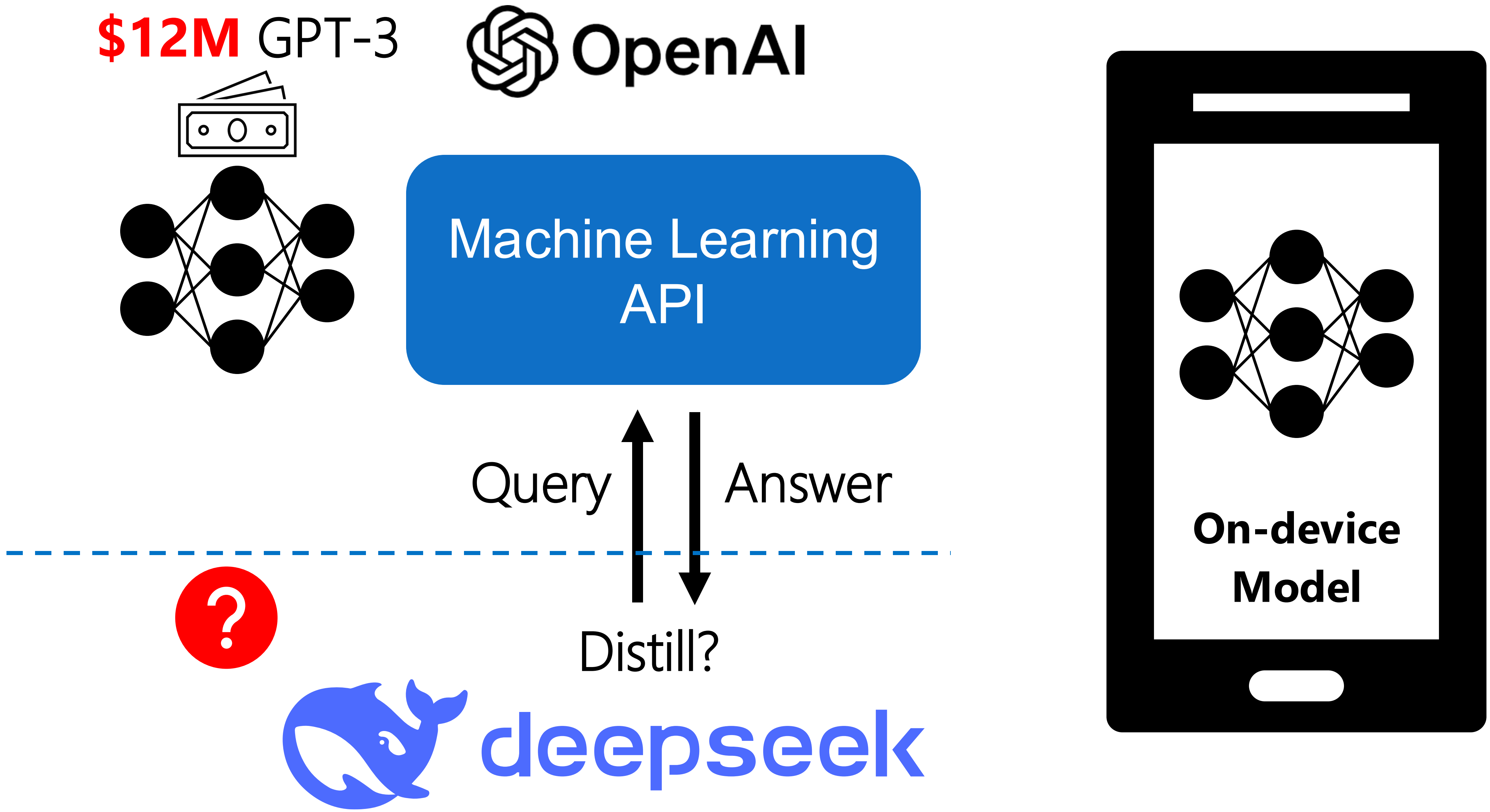
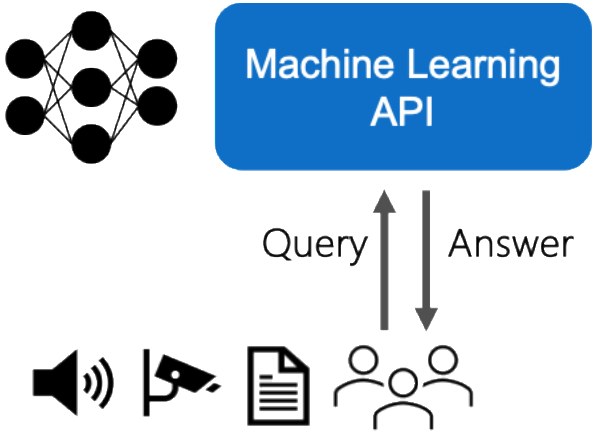


adam-dziedzic.com
adam.dziedzic@cispa.de

Thank You!



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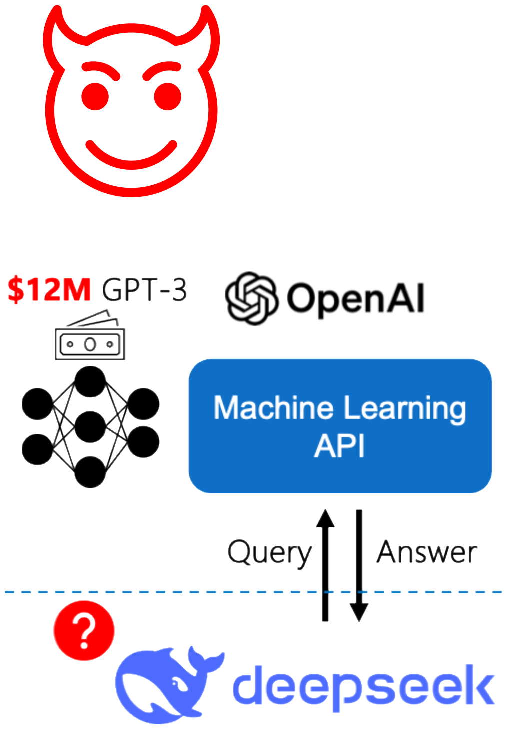

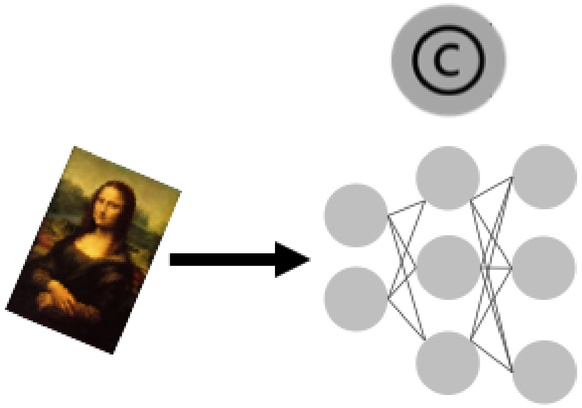

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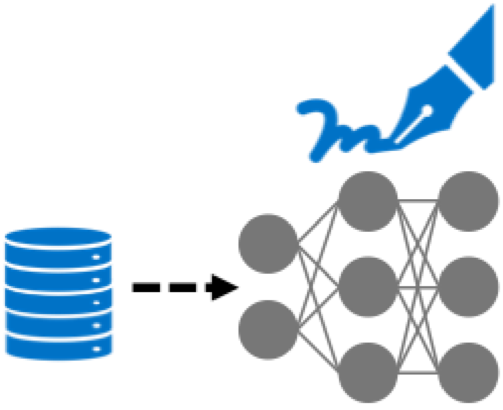

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